

**Prepapig: An Ai-powered Pig Growth Tracker with
Vaccination Scheduling and Pig
Feed Monitoring System**

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with Specialization Track in Business Analytics**

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APPROVAL SHEET

This Capstone Project titled **Prepapig: An Ai-powered Pig Growth Tracker with Vaccination Scheduling and Pig Feed Monitoring System**, prepared and submitted by **Isabella Grace M. Elola, Maris N. De Lunas, Mark Angelo G. Patal** in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology - Major in Business Analytics** is hereby accepted.

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1. INTRODUCTION

This chapter presents the background of the study, the objectives, the significance, and the scope and delimitation.

1.1 Background of the Study

Livestock production plays an important role in ensuring food supply and sustaining the livelihoods of farmers in many parts of the world. Among the various livestock industries, swine production remains a major source of animal protein and agricultural income. As the demand for pork continues to increase, farmers must maintain efficient management practices to ensure healthy livestock development, consistent productivity, and sustainable farm operations. At ANCY's Pig Farm, daily operations involve feeding pigs, checking their condition, and maintaining livestock information. The farm operates under a breeding-based system where all of their pigs are raised on their own farm, allowing full control over the pig growth performance cycle from birth until pigs are ready for market. Pigs are typically raised for four to five months before being sold alive, depending on their weight and condition. Farm caretakers monitor pig development based on observation, ensure proper feeding, and verify that vaccinations are administered according to schedule. These activities are documented manually using notebooks or logbooks, while a calendar is used to track important schedules such as vaccination dates. This manual and paper-based approach represents the current system used by the ANCY's farm to manage its operations.

Despite supporting basic record-keeping, the current system presents several limitations and challenges. Data organization becomes difficult due to the manual recording of information on pig growth performance, feeding activities, and vaccination

schedules. Multiple notebooks and logbooks are reviewed by the caretaker whenever they need to check the pigs' history records. As the number of pigs increases, managing schedules through a calendar becomes increasingly difficult, and important entries may be overlooked, causing caretakers to forget to administer vaccinations to the pigs.

These limitations lead to several operational effects that impact farm performance and decision-making. Reaching desired market weight as monitoring growth performance becomes increasingly difficult, especially for pigs that experience slower development and longer feeding periods due to missed vaccinations. Since feeding is done regularly, feed consumption becomes one of the major expenses in farm operations. However, due to manual recording, feed usage and expenses are often not tracked and only estimated, which makes the actual cost of production and farm profitability assessment difficult to determine. In addition, when vaccination schedules are missed and not properly monitored, some pigs may become weak or more prone to illness, which affects their appetite and feeding behaviour. As a result, these pigs are often separated from the others, further increasing the workload and affecting overall farm efficiency.

Addressing these challenges requires a more systematic and technology-supported approach to livestock monitoring and farm management. According to Wang et al. (2022), AI-based precision pig farming systems and intelligent monitoring technologies can support real-time growth tracking, feeding management, and farm productivity, which improves operational efficiency and livestock management. The integration of AI in the proposed system is justified as it enables data-driven insights, such as feed forecasting and growth trend analysis, and performance monitoring, which is difficult to achieve using manual methods and simple record-keeping systems.

The proposed system, PrepAPig, is designed to provide a centralized digital platform for livestock monitoring at ANCY's Pig Farm. By organizing pig records, feeding information, and vaccination schedules in a single system while supporting feed consumption tracking, feed forecasting, growth trend analysis, and performance monitoring insights, the platform aims to improve farm record management, enhance operational efficiency, support expense and profit analysis, and provide better oversight of livestock performance.

1.2 Objectives of the Study

This section outlines the general and specific objectives that the study aims to achieve in addressing the identified problems.

General Objectives

The main goal of this study is to develop PrepAPig, an AI-powered livestock monitoring and analytics system that assists ANCY's Pig Farm in tracking pig growth performance, managing vaccination schedules, monitoring feed consumption, and analyzing data through a centralized progressive web-based application dashboard. Specifically, this study aims to:

1. Develop a progressive web app that:

1.1. Tracks pig growth performance

1.2. Monitors feed consumption

1.3. Records and manages capital-related expenses, including:

1.3.1. Feeds

1.3.2. Vaccines

2. Manage vaccination and feed scheduling by:

2.1. Automating feed schedules by batch and days

2.2. Managing vaccination schedules and reminders

3. Generate farm performance analysis and reports, including:

3.1. Feed consumption reports

3. 2. Profit analysis

1.3 Scope and Delimitation

This study focuses on the design and development of PrepAPig, an AI-powered pig growth performance tracking system with vaccination scheduling and feed consumption monitoring, specifically implemented for ANCY's Pig Farm. The system is developed as a progressive web application (PWA) that centralizes livestock data and supports farm management through digital monitoring and reporting tools. It includes functionalities such as pig growth performance monitoring, where pig weight and development are recorded and tracked per batch; feed consumption management, which logs feeding data and generates summaries for efficiency evaluation; and vaccination scheduling and reminders to ensure timely administration of vaccines. Additionally, the system provides feed scheduling automation based on batch grouping and pig age, expense recording for feeds, vaccines, and piglets, and data analytics and reporting features, including feed consumption summaries and basic profit analysis to support decision-making.

The artificial intelligence component of the system is clearly defined and limited to growth trend analysis based on recorded weight data, feed consumption pattern analysis, and basic feed forecasting using historical records. The AI implementation is rule-based and data-driven, designed only to provide supportive insights, and does not include advanced machine learning techniques such as computer vision, real-time IoT monitoring, or autonomous decision-making capabilities.

This study is subject to several delimitations. The system is exclusively designed for ANCY's Pig Farm and may require modifications for use in other farms with different operational practices. The farm only operates through a breeding-based system and farrowing system operation, where all pigs are produced from its own breeding stock rather than being purchased from external sources. Data input is manually performed by farm caretakers; therefore, the accuracy of the system depends on consistent and correct data entry. The AI component does not utilize real-time data or external sensors and relies solely on historical data within the system. Furthermore, the system does not include IoT-based automated feeding systems, sensor-based health monitoring such as temperature or activity tracking, or image recognition and disease detection features. Financial analysis is limited to basic expense tracking and profit estimation and does not cover advanced accounting processes. The application requires internet access for full functionality as a progressive web application. Lastly, the system focuses only on growth tracking, feeding, and vaccination management, and does not include other aspects of livestock management such as breeding, genetic analysis, or disease diagnosis.

1.4 Significance of the Study

This study aims to demonstrate the significance of implementing an AI-powered livestock management system in improving operational efficiency, supporting data-driven decision-making, and providing valuable benefits to various stakeholders. By integrating artificial intelligence and data analytics, the system enhances farm management through centralized monitoring, intelligent feed management, automated vaccination scheduling, promoting efficiency and sustainability in livestock management. This study will benefit the following stakeholders:

Farmers and Farm Owners. Farmers and farm owners are provided with access to centralized digital platform that facilitates the monitoring of growth management, feed monitoring and scheduling of vaccination. The integration of AI-powered analytics supports real-time, growth pattern analysis and feed consumption monitoring and feed forecasting, which can enhance the overall productivity and profitability of the farm.

Farmer's Caretakers. Caretakers will benefit from a structured and user-friendly system that simplifies daily farm operation. Automated scheduling and digital record management eliminate the need for multiple notebooks and manual tracking methods. This improves task organization, ensures timely execution of vaccinations and feeding activities.

Livestock Industry. The livestock sector will benefit from the adaption of AI-driven monitoring systems that modernize traditional farming practices. The system demonstrates the application of predictive analytics and data-driven insights in improving production efficiency, feed management, and livestock monitoring. This contributes to the advancement of smart agriculture and promote sustainable and technology-enable livestock operation.

Academic Institution and Students. Academic Institution and Students benefit from a real-word application of system development within the field of information technology and Business Analytics. The study serves as a model for system, design, implementation, and evaluation, enhancing learning in data driven, and technology-based solutions.

Consumers and the general public. Will benefit indirectly from improved livestock management practices that result in more efficient production processes. This

contributes to higher-quality pork products, improved food supply stability, and more sustainable agricultural operations.

Future Researchers and Developers. Researchers and future system developers gain a relevant reference for exploring AI applications in agriculture and livestock monitoring. The study provides a practical framework for integrating analytics, automation, and intelligence system into farm management.

Based on the study, the following Sustainable Development Goals (SDGs) are most aligned with the study. The study supports SDG 2: Zero Hunger by improving livestock productivity and farm management, which contributes to more stable food production and strengthened food security. In addition, it aligns with SDG 9: Industry, Innovation, and Infrastructure by promoting the use of artificial intelligence, digital systems, data analytics to modernize livestock farming and encourage technological innovation in agriculture. Furthermore, it contributes to SDG 12: Responsible Consumption and Production by enhancing feed management, reducing waste, and optimizing production processes to ensure more efficient and sustainable resource use. The study also supports SDG 8: Decent Work and Economic Growth by improving farm productivity, profitability, workflow efficiency for farmers and caretakers. Lastly, it aligns with SDG 4: Quality Education by providing a real-world application of system development that enhances learning opportunities for academic institutions and students in the field of information technology, analytics, and smart farming technologies.

1.5 Definition of Terms

To have a better understanding of the system, the following terms are defined:

Artificial Intelligence (AI): Artificial Intelligence refers to the science and engineering that focuses on developing intelligent systems within the field of computer science domain, which in this system is used as computer-based methods for analyzing data.

AI - Powered: AI - Powered refers to technology that incorporates artificial intelligence (AI) capabilities to perform tasks and respond to complex inputs, helping to monitor pig growth performance and supporting farmers in making timely decisions to improve pig welfare.

Livestock Production: Livestock Production refers to the agricultural practices of raising animals, food, and income serving as the foundation of farm operations, including the management of animal breeding, feeding, health care, and growth to ensure consistency.

Feed Forecasting: Feed Forecasting refers to the use of data and computational methods to predict the amount of feed pigs will consume over a specific period, providing farmers with efficient planning and cost management in farm operation

Progressive Web Application (PWA): A Progressive Web Application (PWA) refers to the web-based application that combines the features of traditional web applications and native mobile applications to provide a responsive, reliable, and app-like user experience across different devices

Profitability Analytics: Profitability analytics refers to the process of strategic support prices, analyzing income and expenses to determine farm profitability by using historical data and cost analysis to support better financial decision-making and management.

Swine Production: Swine Production is the process of raising pigs for commercial purposes, especially for pork production, which involves feeding, monitoring growth, and caring for pigs from birth until they reach market weight to ensure a steady supply of pork and farm income through an organized farming system.

2. LITERATURE REVIEW

2.1 Related System

This section reviews existing systems relevant to the study to identify their features, capabilities, and limitations.

A systematic review by Gómez et al. (2021) describes how sensors, automation, and AI support pig farming by tracking growth, feed intake, environmental conditions, and health. Their systems include wearable sensors, automated weighing, and data analytics, which facilitate early detection of health problems and improvement of animal welfare. A significant limitation is the high cost and complexity of the required equipment, rendering these technologies inaccessible to smaller farms. The proposed system overcomes these challenges by offering a simpler, web-based alternative that integrates growth tracking, feed management, vaccination scheduling, and basic financial analysis into a single platform, thereby eliminating the need for costly hardware and multiple disconnected systems.

In addition, a technique developed by Misiura et al. (2021) forecasts pig growth using feed consumption and past performance data, applying it to their system. This approach assists farmers in optimising feeding strategies and production scheduling by providing predictions based on historical records. However, the system is confined to feeding and weight estimation only. It does not include vaccination tracking, financial monitoring, or centralised record-keeping, nor does it address how farmers can combine growth forecasting with other essential farm operations. Unlike existing systems, PrepAPig merges growth monitoring with feeding plans, vaccination timelines, and basic financial

evaluation in one integrated system, thereby allowing farmers to observe weight trends, feeding adjustments, and vaccination events together on a single dashboard.

Another relevant study, that of Van Klompenburg and Kassahun (2022), examines the application of data-driven techniques and machine learning to improve decision-making in pig farming. Their analysis covers feed intake, growth rates, and environmental factors, demonstrating significant potential for transforming farm management through predictive models and performance indicators. Although the findings are promising, many of the methods remain experimental and have not been integrated into everyday farm systems. These methods lack connections to vaccination management, expense monitoring, and user-friendly dashboards that would enable farmers to apply insights directly. To bridge these gaps, PrepAPig transforms these data-driven ideas into a functional web platform that integrates analytics with daily farm management tasks such as vaccination scheduling, expense tracking, and reporting, thereby making advanced decision support accessible to non-technical users.

Similarly, research by Aquilani et al. (2022) states that livestock monitoring is a fundamental component of Precision Livestock Farming because it enables continuous, real-time, and individual-level observation of animals through technologies comprising sensors, RFID systems, GPS, and automated data collection tools. These technologies improve farm management by providing objective and continuous information on animal behaviour, health status, and feed intake, thereby supporting growth performance. However, the study focuses on data collection technologies rather than on how the collected data can be integrated with active management functions such as vaccination scheduling, feed planning, or financial tracking. The systems described are primarily

monitoring-oriented and do not offer automated alerts for upcoming vaccinations or profit analysis based on feed consumption. The proposed platform improves upon these limitations through AI-based monitoring that enables continuous tracking of feed consumption and growth performance data while supporting automated vaccination scheduling based on health and performance indicators, thus bridging the gap between passive observation and active farm management.

Moving to scheduling and automation, a study by Berckmans (2021) highlights precision pig farming driven by AI, automation, computer vision, and predictive analytics. Features include automated feeding systems, behaviour tracking, and real-time health alerts, all designed to increase productivity and reduce labour requirements. Despite these advances, most solutions are designed exclusively for large, well-capitalised farms and are not suitable for smaller operations that continue to rely on manual methods. Each system typically performs only one function, automated feeding, behaviour tracking, or health monitoring, rather than managing the entire farm, which forces farmers to coordinate multiple disconnected tools. The proposed system overcomes these challenges by providing an all-in-one platform that handles pig tracking, feeding schedules, vaccination reminders, and financial oversight, thereby making integrated farm management accessible to farms of all sizes.

A simulation system built by Pham et al. (2021) investigates precision feeding techniques. Their system models how different feeding methods affect pig growth, which is valuable for researchers studying feed allocation strategies before applying changes in real farms. Nevertheless, the system is intended purely for research and simulation, not for real-time agricultural tasks. Farmers cannot use it to manage daily operations such as

recording feed intake, adjusting portions based on weight, or tracking vaccination events, and it lacks expense tracking and farm documentation. The proposed system overcomes these challenges by providing a practical web application designed specifically for everyday farm management, thereby translating research insights into actionable tools for feeding, tracking, and scheduling.

A review by Koley et al. (2022) examines how adjustments in feed composition and timing can improve growth and reduce costs. Their emphasis on data and structured feeding programs provides a solid theoretical foundation for precision feeding. However, the study is largely theoretical and does not offer farmers any actual software or tools to implement these strategies. It also focuses exclusively on feeding, ignoring health monitoring, vaccination planning, and financial management, thereby leaving a gap between academic recommendations and practical application. To bridge these gaps, PrepAPig integrates feed tracking and forecasting into a complete farm management system that also covers growth monitoring, vaccination management, and financial analysis, thus turning theoretical feeding strategies into actionable, integrated tools.

An automated system developed by Tangka et al. (2021) manages feed, water, and sanitation on farms. By automating these basic tasks, the system reduces manual labour and improves resource use, assisting farms in streamlining daily chores. However, its functionality ends there. It cannot perform growth tracking, vaccination scheduling, financial analysis, or decision support. The study does not address how automation can be extended to other critical farm operations such as health monitoring or cost analysis. The proposed platform improves upon these limitations through combining monitoring, scheduling, and data analytics into a single, easy-to-use platform that extends automation

to feeding, vaccination, and financial tracking, thereby providing a holistic solution rather than isolated automation.

A conceptual framework presented by Marjuki and Wittayakun (2021) proposes structured feeding schedules and data-based decision-making to improve livestock production and reduce waste. Their work highlights the importance of organised feeding programs as a cornerstone of efficient farm management. Nevertheless, the work remains a concept and does not provide a working system for farmers to use. It also fails to link feeding management to health tracking or financial record-keeping, leaving the ideas abstract and difficult to operationalise. Unlike existing systems, PrepAPig delivers a functional web-based system that handles feeding, growth tracking, and vaccination management together, thereby turning theoretical concepts into practical farm management tools that are accessible and user-friendly.

Concerning farm performance analysis and reporting, a review by Melak et al. (2024) demonstrates that AI can reduce manual labour and improve efficiency in livestock farming through continuous monitoring of animal behaviour, feeding patterns, and farm conditions. Their analysis highlights AI-assisted data analysis as a means to achieve better farm management, covering capabilities such as health monitoring and vaccination management. However, because the study is a review, it does not propose how to integrate these analytical capabilities into a single, practical system for daily use. As a result, functions such as monitoring, scheduling, and reporting remain fragmented across separate discussions, with no unified platform that combines them. The proposed system overcomes these challenges by providing a unified interface that combines monitoring, scheduling,

and reporting into one integrated platform, allowing farmers to access all AI-driven insights without switching between tools.

Smart agri-systems are described by Collins and Smith (2022) as integrating various data sources, including sensor networks (cameras, load cells, RFID), automated data capture, and digital twinning, to provide a comprehensive view of livestock production. This integration permits farmers to consider multiple variables simultaneously, potentially improving decision-making. While this integration is powerful, it does not guarantee usability. Many such systems are technically advanced and require specialised knowledge to interpret the data, creating a barrier for smaller or less technologically equipped farms. Farmers may struggle to comprehend dashboards, configure alerts, or derive actionable insights. To bridge these gaps, PrepAPig addresses this by balancing technical sophistication with simplicity, ensuring that users can easily understand and apply the information through an intuitive web-based dashboard, thereby lowering the technical barrier to entry.

Digital livestock farming is the focus of Neethirajan (2021), where manual record-keeping is replaced by automated data management. Digital systems enable continuous collection of data on animal health, behaviour, and environmental conditions using biometric sensors (accelerometers, thermometers, RFID), acoustic monitoring for cough detection, and big data analytics for real-time alerts. These technologies provide more accurate and consistent records than manual methods. Nevertheless, digitalisation introduces challenges such as high costs, data integration issues, and the need for reliable infrastructure. Farmers may receive large amounts of data without clear guidance on how to interpret it for daily decisions. The proposed platform improves upon these limitations

through not only collecting data but also presenting it in a clear, actionable form through a user-friendly interface, focusing on what farmers need to know rather than simply providing raw numerical outputs.

A broad overview of AI applications in livestock farming is provided by Distant et al. (2025), covering tasks such as behaviour analysis, recognition, and automated decision-making. Their review includes animal identification, abnormality detection, and behaviour recognition using various AI techniques. AI clearly supports continuous monitoring and reduces manual intervention. Yet, a recurring problem like fragmenting, most systems are built for specific problems, resulting in separate tools for monitoring, analysis, and management. This lack of interoperability creates inefficiencies, forcing users to switch between systems to complete different tasks such as tracking growth, scheduling vaccinations, and analysing profits. Unlike existing systems, PrepAPig solves this fragmentation by providing a unified platform that integrates growth monitoring, vaccination scheduling, feed consumption tracking, expense recording, and profit analysis into a single, cohesive system, thereby eliminating the need for multiple tools and reducing operational complexity.

The reviewed literature reveals several recurring limitations. Numerous existing systems depend on expensive hardware components, including sensors, radio-frequency identification (RFID) tags, and automated weighing stations, which renders them inaccessible to small-scale and medium-scale farms. The majority of current solutions focus on a single operational function, such as feeding, growth monitoring, or health tracking, without incorporating vaccination scheduling, expense recording, or profit analysis. Furthermore, certain studies remain theoretical or simulation-based and do not

provide deployable software for routine farm management. Among digital systems, fragmentation persists, requiring farmers to utilise multiple disconnected tools, which introduces inefficiencies and data silos. The proposed system, PrepAPig, addresses these gaps by offering an efficient, web-based, unified platform that requires no specialised hardware. It integrates growth tracking, feed management, vaccination scheduling, and financial analysis, delivering practical insights through an intuitive dashboard designed for farms of all sizes.

2.2 Related Technologies

This section discusses the technologies and frameworks used in the development of the proposed system and explains their relevance to the study.

Progressive Web Application (PWA)

According to Cherukuri (2024), Progressive Web Apps (PWAs) combine service workers, web app manifests, and modern JavaScript frameworks to deliver app-like experiences with offline capability, push notifications, and background synchronization. These features operate within a web browser without requiring installation from an app store.

Development of PrepAPig as a PWA provides farm caretakers with an installable, offline-first application that functions reliably even during intermittent internet connectivity. Service workers cache essential farm data, including pig batches, feed logs, and vaccination schedules, thereby enabling offline access and data entry when the network is unavailable. Push notifications remind users of upcoming vaccinations, feeding schedules, and overdue tasks. Background sync automatically synchronizes offline entries once connectivity is restored. By adopting PWA technology, PrepAPig offers a native-app

experience, reduces dependence on constant internet access, and improves user engagement and operational efficiency on both mobile and desktop devices.

React

According to the research conducted by Piastou (2023), a comparative assessment of React, Angular, and Vue.js shows that React provides efficient rendering and scalability due to its virtual DOM, while Angular suffers from longer initial load times and higher memory usage, making React a more suitable choice for a progressive web application. React's virtual DOM minimises DOM manipulation, and its extensive ecosystem provides mature libraries for offline caching (Workbox), state management (Context API), and component reusability.

For building interactive user interfaces, React is employed in PrepAPig to create dynamic dashboards for pig growth tracking, feed consumption logs, vaccination scheduling, and profit analysis. React's component-based architecture allows the development of reusable UI components such as data entry forms and charts, which can be easily maintained and extended as farm management needs evolve. The virtual DOM ensures that only changed elements are updated, reducing page reloads and providing a smooth, app-like experience even on mobile devices with limited processing power. Additionally, React's compatibility with offline caching libraries (e.g., Workbox) and state management tools supports PrepAPig's offline-first PWA requirements, enabling farm caretakers to record data even without an internet connection and synchronize later.

MySQL

MySQL was selected because it is lightweight, open-source, offers strong data protection, multi-user access, and runs on low hardware specifications (Wahyudi et al.,

2022). Its ACID compliance and cost-free deployment make it ideal for a farm management system. Using this technique, all farm-related data are stored in MySQL within PrepAPig, including pig batches, feed consumption logs, vaccination schedules, expense records, and user accounts. MySQL's ACID compliance ensures that financial and health records remain consistent, while its support for complex SQL queries enables the generation of feed consumption reports, growth trend summaries, and profit analysis. Additionally, MySQL's low resource requirements and open-source licensing make it an affordable and scalable choice for a farm management PWA intended for long-term use.

Node.js for Backend Runtime

According to Basumatary and Agnihotri (2022), Node.js is a cross-platform JavaScript runtime that uses an event-driven, non-blocking I/O model to handle multiple concurrent requests efficiently with a single thread.

Node.js processes API requests, manages MySQL database operations, handles push notifications for vaccination reminders, and coordinates offline data synchronization within PrepAPig. Its non-blocking nature allows the system to serve multiple farm caretakers simultaneously without performance degradation, while using JavaScript across both frontend and backend simplifies development and maintenance.

Express.js for Backend Web Frame

According to Kostrzewa and Milosz (2024), Express.js is a minimalistic and flexible web framework built on Node.js, optimized for handling multiple concurrent queries with better performance than ElysiaJS in parallel scenarios.

Express.js creates RESTful API routes for pig batch management, feed consumption logs, vaccination schedules, expense tracking, and report generation in

PrepAPig. It also handles request routing, middleware implementation for authentication and validation, error handling, and coordination between the React frontend and MySQL database. Its non-blocking, event-driven nature allows the system to serve multiple farm caretakers simultaneously without performance degradation, while its minimalistic design keeps the backend modular, maintainable, and easy to extend as farm management needs evolve.

Tailwind CSS

According to Al Salmi (2023), Tailwind CSS is a utility-first CSS framework that provides developers with full control over customization without relying on pre-designed components. It uses low-level utility classes to build unique interfaces quickly, and its PurgeCSS tool removes unused styles, resulting in a very small final CSS file size.

Tailwind CSS styles the entire Progressive Web App, including the login page, pig growth dashboards, feed consumption forms, vaccination calendar, expense tracking tables, and profit analysis charts. Its utility-first approach allows rapid development of a consistent, responsive layout that works seamlessly on desktops, tablets, and mobile phones. Because Tailwind does not impose predefined themes, the system achieves a clean, professional look tailored specifically to farm management needs, while the small CSS footprint ensures fast loading times even on low-end devices and unreliable network connections.

Firebase Cloud Messaging

According to Putro et al. (2023), Firebase Cloud Messaging (FCM) enables web applications to send push notifications reliably over HTTPS, delivering messages to users even when the browser is closed or the application is not actively open. FCM eliminates

the need for users to keep the software open to receive updates, as notifications appear on the device regardless of the application's state.

FCM sends timely alerts for vaccination schedules, feeding reminders, and overdue tasks directly to farm caretakers' mobile devices or desktop browsers. FCM's cross-platform support ensures notifications reach Android, iOS, and web clients consistently. To complement offline functionality, the system also implements service workers (via PWA) to cache static assets, but the primary notification mechanism is FCM. This approach guarantees that critical farm reminders are delivered promptly without requiring users to keep the application open, improving responsiveness and reducing the risk of missed vaccinations or feeding events.

Vercel

According to Ramasamy et al. (2024), Vercel is a cloud platform optimised for Next.js applications, offering GitHub integration for CI/CD, secure environment variable management, a global CDN, and automatic scalability. Vercel was selected for deploying PrepAPig because it provides seamless scalability, allowing the application to handle increased traffic during peak farm management periods without manual intervention. Its deployment efficiency is enhanced through automatic builds and preview deployments on every code push, reducing the risk of errors and accelerating release cycles. As a web hosting platform, Vercel offers global edge caching, which minimises latency for farm caretakers in different geographic locations, and built-in support for PWAs and service workers, aligning with PrepAPig's offline requirements. The platform's secure environment variable management also protects sensitive API keys and database credentials. Deployment on Vercel brings automatic deployments, efficient handling of

static and server-rendered pages, global edge caching, and secure storage of API keys. Vercel's built-in support for PWAs and service workers aligns with PrepAPig's offline requirements, ensuring fast load times regardless of user location.

Chart.js

According to Benbba (2021), Chart.js is a free, open-source JavaScript library that uses HTML5 Canvas to render data visualisations. It offers a simple, configuration-based approach with built-in tooltips, legends, and animations, requiring minimal coding.

Chart.js fulfills data visualisation needs for growth trends, consumption comparisons, and profit distribution. Its ease of use enables rapid dashboard development that farm caretakers can understand quickly, while canvas-based rendering performs well on mobile devices. Built-in tooltips and hover interactions provide detailed insights without cluttering the interface. For PrepAPig's descriptive analytics, Chart.js balances simplicity, functionality, and performance effectively.

HTML5

According to Rahman (2024), HTML5 introduces native support for semantic elements such as header, footer, nav, and section, new input types including date, time, email, url, and search, form validation attributes like required and pattern, offline application cache, and the canvas element for dynamic graphics. These features enable rich, interactive web applications without third-party plugins, improve accessibility, and support cross-platform deployment on both desktop and mobile devices.

PrepAPig, a Progressive Web App, uses HTML5 as its core markup language to structure farm-management interfaces, including growth dashboards, feed logs, vaccination calendars, expense tables, and profit charts. It employs the <canvas> element

for basic data visualizations and semantic tags (<header>, <nav>, <section>, <footer>) for layout organization. HTML5's built-in form validation (such as required and pattern attributes) ensures accurate data entry for pig weights, dates, and feed amounts, while the offline application cache supports offline-first functionality. By leveraging native HTML5 features, PrepAPig reduces reliance on external plugins and delivers a consistent, responsive user experience across devices.

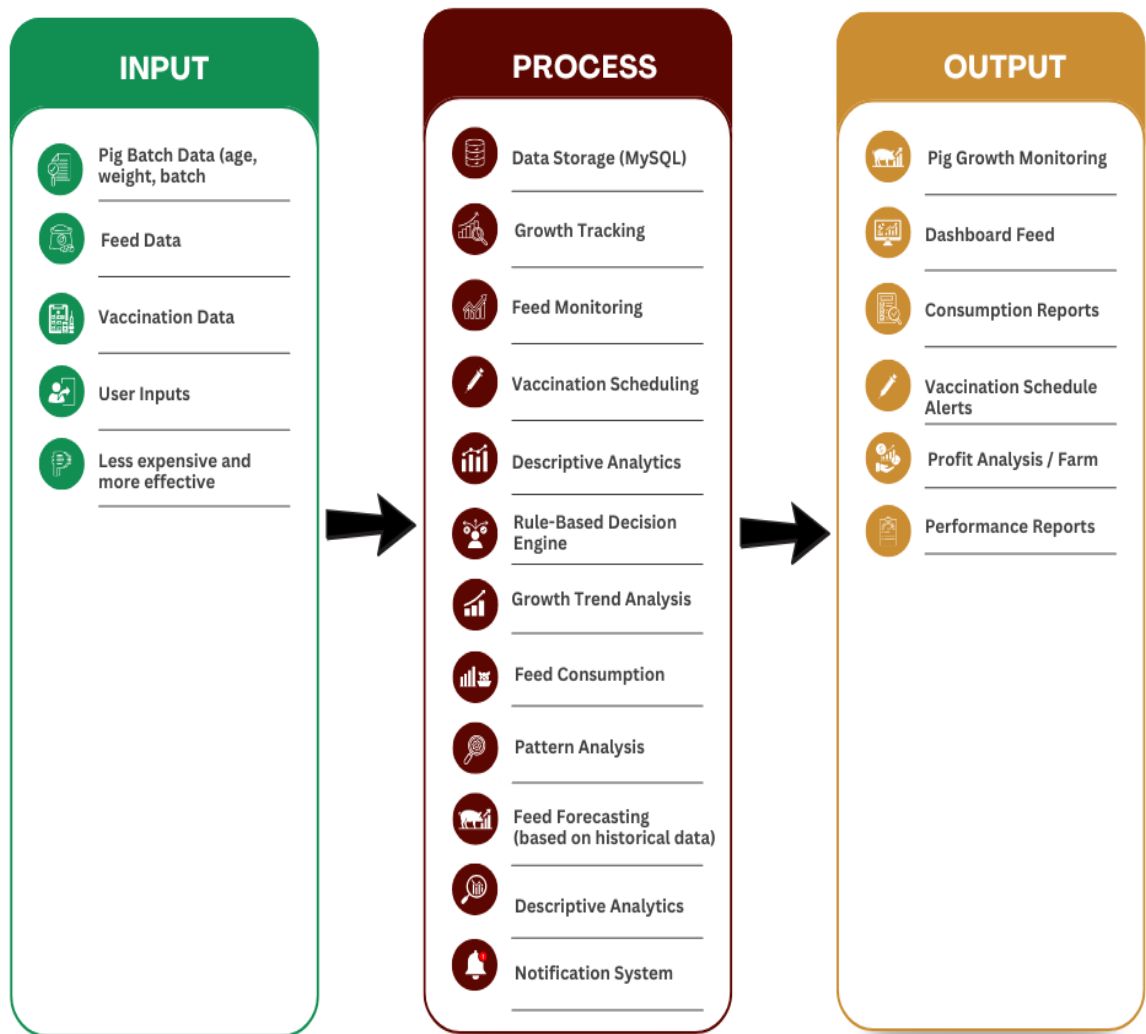
The technologies selected for PrepAPig form a cohesive, offline-first ecosystem tailored to the operational realities of farm management. React's component-based UI and virtual DOM work in tandem with Tailwind CSS to deliver fast, responsive dashboards, while the PWA architecture (service workers, web app manifest) ensures that critical data entry and retrieval remain possible during network interruptions. Node.js and Express.js provide a lightweight, non-blocking backend that handles concurrent requests, synchronizes offline changes via background sync, and sends timely push notifications through Firebase Cloud Messaging. MySQL offers ACID-compliant storage for farm records, and Chart.js translates this data into actionable visualizations. Deployed on Vercel with global edge caching and CI/CD, the entire system remains scalable and maintainable. Together, these technologies address the core requirements of reliability, accessibility, and efficiency for small-scale pig farming.

2.3 Conceptual Framework

The study adopts an Input-Process-Output (IPO) model as the conceptual framework guiding the design and development of the PrepAPig system. This model presents a structured representation of how inputs are processed by the system to produce outputs that support efficient pig farm management and monitoring. The framework

illustrates the flow of data, the interaction of users and external services, and the system processes involved in managing pig growth tracking, feed monitoring, vaccination schedules, and analytics for improved farm decision-making.

Figure 1.
Conceptual Framework



The process begins with the input model, which serves as the foundation of the system. This layer consists of all essential data required for system operation, including pig batch information such as age, weight, feed data, vaccination records, user inputs, and it. These datasets are carefully collected to ensure accuracy and reliability. Additionally, the

system emphasizes cost-effectiveness by utilizing accessible and efficient data inputs. At this stage, all raw data is prepared and structured before entering the system for processing.

The next stage is the process model, which represents the core functionality of PrepApig. In this phase, all input data is stored, organized, and managed within a centralized database system using MySQL. The system performs multiple analytical and operational functions, including growth tracking, feed monitoring, and vaccination scheduling. It also incorporates advanced analytical features such as descriptive analysis, growth trend analysis, feed consumption analysis, and pattern recognition. A rule-based decision engine is utilized to interpret data and generate recommendations, while forecasting techniques use historical data to predict future feed requirements. Furthermore, an integrated notification system ensures that users receive timely alerts regarding vaccination schedules and other important farm activities. Through these interconnected processes, raw data is systematically transformed into actionable insights that support informed and intelligent decision-making.

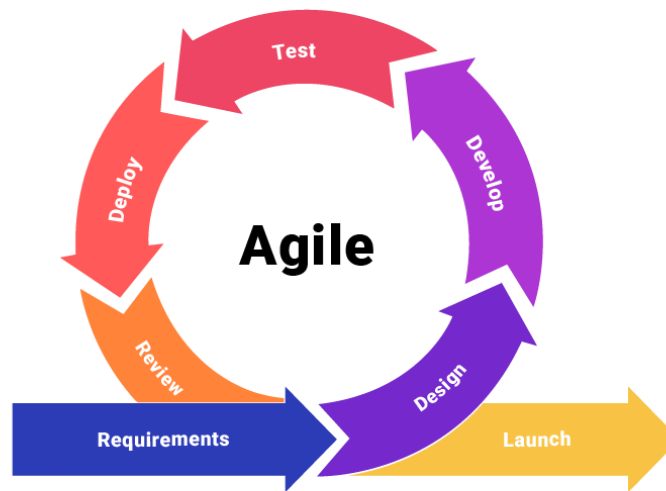
Finally, the output model presents the processed information in a user-friendly and accessible format. This includes pig growth monitoring dashboard, feed visualization tools, consumption and performance reports, vaccination alerts, and profit analysis. These outputs are designed to simplify complex data that will allow the farmers to easily interpret results and apply them to real-world farm management decisions. As a result, the system will enhance efficiency in feeding strategies that improve pig growth performance.

3. METHODOLOGY

3.1 Research Design

This section describes the development approach used in the study. It explains how the system will be designed, configured, simulated, and evaluated, including the development phases, client involvement, and iteration process.

Figure 2.
Agile Methodology Framework



The study plans to utilize Agile Development Methodology in developing PrepAPig, a progressive web-based livestock monitoring and analytics system. Agile will be used because it supports iterative and flexible development process, allowing continuous improvements (Zaveri et al., 2024). This approach is applicable to the study as the study requires multiple modules such as pig growth tracking, feed monitoring, vaccination scheduling, and analytics reporting.

The system is intended for farm owners and caretakers for them to manage pig records, monitor growth performance, track feed consumption, schedule vaccinations, and generate reports. Existing records from ANCY'S pig farm will be encoded into the system as baseline data, new pig batches or individual pig records can also be added through a validated digital form to ensure data accuracy and prevent duplicates.

The system will include modules for pig growth monitoring, feed management, vaccination scheduling, and analytics reporting. Pig records will be organized by batch or individual to make monitoring more structured and efficient. The scheduling module will provide interface for tracking vaccinations dates, feeding schedules, and other farm-related tasks. Automated reminders will also be included to help reduce missed schedules and improve daily task management.

In addition, the system will integrate AI-assisted growth trend analysis and feed forecasting features to support farm decision making. The system will analyze pig weight records, feed consumption data, and growth history to identify development patterns, estimate food requirements, and generate data driven insights for improving productivity and farm profitability. This feature will help farm owners make more informed management decisions based on recorded farm data.

System evaluation will be conducted with the farm owner and caretakers of ANCY's Pig Farm to assess the system's functionality, usability, efficiency, and effectiveness compared to the existing process of monitoring and record keeping.

PHASE 1 – Requirements Gathering

The requirements gathering phase involves identifying and documenting the functional and non-functional requirements of the system. Data will be collected through

interviews, research, and review of existing farm records involving the farm owner and the caretakers of ANCY's Pig Farm. The client (farm owner and caretakers) will be actively involved in this phase by providing operational needs, existing records, and feedback on required features. Client review meetings will be conducted to validate and prioritize the gathered requirements. Key requirements will include pig growth monitoring, feed consumption recording and forecasting, vaccination scheduling, and analytics reporting.

PHASE 2 – Design

The design phase involves the overall system architecture, database structure, and user interface design, which will be created based on the gathered requirements. This includes the design for the Progressive Web App (PWA) layout, pig monitoring dashboard, feed management module, vaccination scheduling interface, and analytics reporting interface. Wireframes, mockups, and process diagrams will also be developed to ensure the usability, accessibility, and alignment with the daily operational need for farm users. Prototype evaluation will be conducted with the client to review wireframes and mockups before finalizing the design. Client feedback on initial design mockups will be sought to confirm that the interface meets user expectations.

PHASE 3 – Iterative Development

The iterative phase involves iterative development using Agile principles. Core features such as pig monitoring feed consumption recording, feed forecasting, vaccination scheduling and report generation will be implemented in incremental development cycles. Each iteration will be focusing on delivering functional modules that can be reviewed, tested, and improved based on the user feedback and observed system performance. The client will be involved at the end of each iteration by reviewing the delivered module and

providing feedback, which will inform the next cycle of development. Feedback validation sessions will be held to confirm that the implemented features align with client expectations. This iterative feedback process will ensure that the system continuously evolves to meet farm operational needs.

PHASE 4 – Testing

The testing phase will involve continuous and iterative testing of the system throughout the entire development process. This will include functionality testing to verify that key system features work properly, usability testing to assess interface clarity and user experience, simulation of system operations using baseline farm data to validate growth trends and feed forecasts, and analytics validation to ensure accuracy of outputs. Issues identified during testing will be documented and addressed in succeeding development iterations to improve system reliability and performance. The client will participate in usability testing and analytics validation by comparing system-generated insights with actual farm records. Client review meetings will also be conducted to discuss testing results, gather feedback, and prioritize fixes for the next iteration.

PHASE 5 – Deployment

The deployment phase ensures that after successful testing the Progressive Web Application platform created for actual use at ANCy's Pig Farm can be accessed effectively through available devices such as smartphones, tablets, or computers. User orientation and system demonstration will also be conducted to ensure that the farm owner and caretakers understand how to use the system for daily farm operations.

PHASE 6 – Review

The review phase will involve gathering feedback from the farm owner and the caretakers to assess system performance and whether the objectives of the study have been met. User acceptance testing and feedback analysis will be conducted to identify the strengths, limitations, and possible areas for enhancements. Client review meetings, feedback validation, and prototype evaluation (if further refinements are needed) will be conducted to ensure all concerns are addressed. Client involvement in this phase will be crucial, as their direct feedback will determine the system's acceptance and guide recommendations for future improvements. Findings from this phase will support the study's conclusion and recommendations while ensuring that the system remains adaptable for future farm operational needs.

3.2.1 Client Profile

This section presents the analysis of the client's current pig farming operations, organizational setup, existing processes, and identified system requirements, which serve as the basis for the design and development of the PrepAPig system.

Description of the Client

The client of this study is ANCY's Pig Farm, a local pig farming enterprise engaged in swine production. The operation is located in a rural agricultural area in Cahigpitan, Calaca City, Batangas, Philippines, and it is considered a small- to medium-scale farm that is directly managed by the owner with the assistance of a limited number of caretakers. The farm started as a small-scale livestock activity with the primary purpose of raising pigs for commercial production and contributing to the local pork supply. Over time, it has maintained its own breeding stock, ensuring that all piglets are produced internally rather

than purchased from external sources. The main activity of the farm includes pig breeding, raising, feeding, vaccination, and health management until the pigs reach marketable weight, to ensure the proper growth and development of livestock.

In terms of organizational structure, ANCY's Pig Farm is classified as a family-owned agricultural enterprise under the livestock production industry. It is typically handled by the farm owner with the assistance of one worker and family members who oversee all farm operations. Due to its limited manpower, the farm relies on traditional farming practices due to limited access to advanced technologies. The farm operates within the agricultural sector, specifically focusing on livestock production and swine production, where production depends entirely on controlled reproduction and internal herd management rather than external sourcing of piglets, allowing the farm to maintain control over production quality and herd development.

Currently, the farm operates using a fully manual system for managing its daily activities. Records such as pig growth performance, feed consumption, vaccination schedules, and breeding information are documented through notebooks and simple files. Feeding practices are based on estimation rather than precise measurement, while monitoring of pig health and vaccination schedules is done manually by the farm owner and caretakers. Daily operations include feeding pigs, checking their health condition, cleaning pens, and tracking breeding cycles. Most decisions are based on the farmer's personal experience and direct observation rather than systematic and data-driven approaches (Funk et al., 2024). However, the current setup has several limitations, including lack of a centralized data management system, risk of human error, delayed record updates, and difficulty in tracking real-time information. The farm also relies only

on basic tools such as handwritten logs and mobile phones, which are insufficient for efficient and organized farm management. These challenges highlight the need for an improved system such as PrepAPig to enhance accuracy and efficiency of the farm management.

3.2.2 Current System Analysis

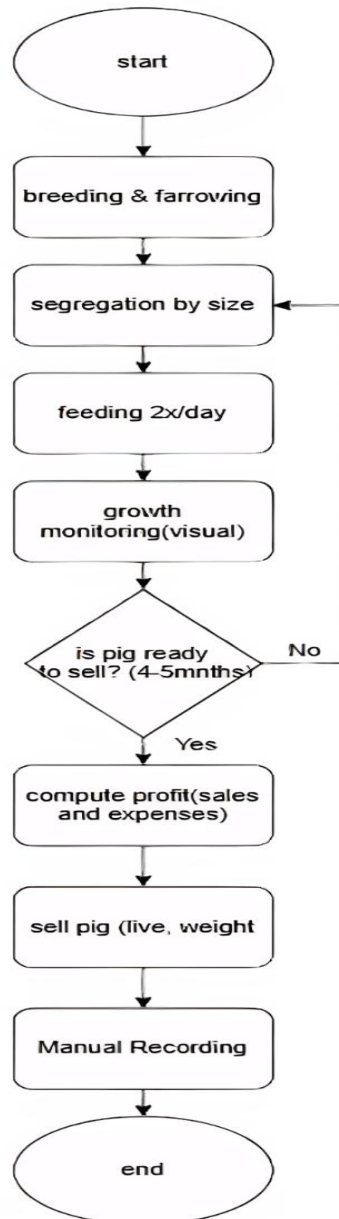
The current system used by the client follows a traditional and manual approach in managing pig farming operations. Most activities are performed through direct observation, estimation, and handwritten documentation instead of using an automated monitoring system. Daily farm management mainly depends on the owner's experience in handling breeding, feeding, monitoring, and selling processes. While this method allows the farm to continue its operations, it creates difficulties in maintaining accurate records, tracking pig development, and efficiently managing farm data.(Marić et al., 2025).

The process begins with breeding and farrowing, where piglets are produced and raised within the farm. After birth, the pigs are segregated according to their size to ensure proper grouping and management. Once grouped, the pigs undergo regular feeding twice a day as part of their daily nutritional routine. During the growing period, the owner performs visual growth monitoring to observe the pigs' condition and development. Since monitoring is done manually, the evaluation of pig readiness for selling is mainly based on physical appearance and estimated growth rather than recorded analytical data.

After monitoring, a decision is made to determine whether the pig is already ready for selling, usually within four to five months. If the pig is not yet ready, it returns to the feeding and monitoring cycle until the desired market weight is achieved. Once the pig is considered ready, the owner manually computes the estimated profit by comparing sales

and expenses. The pigs are then sold based on live weight, and all related information is manually recorded in notebooks or simple files for documentation purposes.

Figure 3.
Process Flow Diagram



The current system analysis shows that the client's pig farming operations rely heavily on manual processes and experience-based management. Activities such as feeding, monitoring pig growth, recording farm data, and evaluating pig readiness for selling are performed through direct observation and handwritten documentation. While this traditional approach allows the farm to operate continuously, it creates challenges in maintaining accurate records, monitoring pig development efficiently, and analyzing farm performance. The findings emphasize the need for a more organized and automated system to improve data management, operational efficiency, and decision-making in pig farming operations.

3.2.3 Problem Identification

Based on the analysis of the current operational setup of the Ancy's pig farming process, the following problems were identified as the primary limitations of the existing manual livestock management system:

1. **Absence of a Centralized Farm Management System.** The client currently has no centralized platform for storing, organizing, and managing farm-related information. Important records such as pig batch data, feed consumption, vaccination schedules, sales, and expenses are manually documented in notebooks or basic files. Since records are scattered across different storage methods, farm data becomes difficult to retrieve, update, and monitor consistently.
2. **Lack of Accurate Pig Growth Monitoring and Analytics.** Pig growth and market readiness are evaluated through manual observation and estimation. The farm has no automated system for tracking growth performance, analyzing records, or

generating growth trends for each pig batch. As a result, selling decisions may become inconsistent due to inaccurate assessment of pig development.

3. **No Systematic Feed Monitoring and Forecasting Mechanism.** Although feeding is conducted regularly, feed allocation and consumption are not properly measured or analyzed. The absence of a structured feed monitoring system makes it difficult to determine feed efficiency, estimate future feed requirements, and identify feed wastage. This may increase operational costs and affect pig growth consistency. ect pig growth consistency.
4. **Manual Monitoring of Vaccination and Health Management Activities.** Vaccination schedules and health monitoring activities are tracked manually through memory or handwritten records. Since the farm has no automated reminder or scheduling system, important vaccination schedules are delayed, overlooked, or inconsistently documented. This increases the risk of improper health management practices that negatively affect pig health, growth performance, and overall farm productivity.
5. **Limited Data Analysis and Decision-Making Capability.** The current setup lacks analytical and reporting tools that can assist the owner in evaluating farm operations effectively. Activities such as computing expenses, estimating profits, and monitoring production performance are manually performed, requiring significant time and effort. The data analysis is not automated, management decisions related to feeding, monitoring, and profitability are based mainly on assumptions and manual calculations instead of accurate operational insights.

6. **Absence of Automated Notifications and Real-Time Monitoring.** The existing system provides no automated notifications, reminders, or real-time monitoring capabilities for farm activities. Tasks such as feeding schedules, growth monitoring, and vaccination tracking depend entirely on manual checking and personal supervision. This increases the number of missed activities, delayed monitoring, and operational inefficiencies, especially when managing multiple pig batches simultaneously.
7. **Difficulty in Generating Accurate Operational Reports.** Since farm records are manually maintained, generating reports related to pig growth, feed consumption, expenses, and sales becomes time-consuming and prone to errors. The owner must manually gather and compute information from different sources before producing summaries of farm performance. This limitation prevents the farm from obtaining timely and accurate reports necessary for effective operational planning and decision-making.

3.2.4 Functional Requirements

This section presents the proposed PrepAPig system, highlighting its key features and functionalities designed to improve pig farming management processes, based on the requirements gathered through consultations with ANCY's Pig Farm to ensure alignment with its operational needs and user roles.

1. OWNER

This section presents the key features and functionalities of the proposed PrepAPig system. The identified requirements were based on consultations with the

client and the analysis of the farm's current operations to ensure that the system meets the needs of efficient pig farm management.

FR-01: Owners are able to manage pig batch records. This includes recording pig batch information such as batch name, number of pigs, age, weight, and monitoring dates.

FR-02: Owners can monitor pig growth through recorded weight and age data to observe growth progress and development trends.

FR-03: Owners can manage feed records by recording feed consumption and monitoring feed usage per batch.

FR-04: The system shall generate feed schedules and reminders based on the pig batch age and feeding stage.

FR-05: Owners are able to manage vaccination records including vaccination schedules, vaccine status, and completed vaccinations.

FR-06: The system shall provide reminders and notifications for upcoming vaccinations, overdue schedules, and feed changes.

FR-07: Owners can manage capital-related expenses, including feeds, vaccines, and piglet costs, for monitoring farm expenditures.

FR-08: Owners can view farm analytics such as growth trends, feed consumption patterns, and feed forecasting summaries.

FR-09: Owners can generate view reports related to pig growth, feed consumption, vaccination records, and farm performance.

FR-10: The system shall support offline access and synchronize recorded data once internet connectivity is restored.

FR-11: Owners are able to securely log in to access the system and manage farm records.

Figure 4.
Use Case Diagram

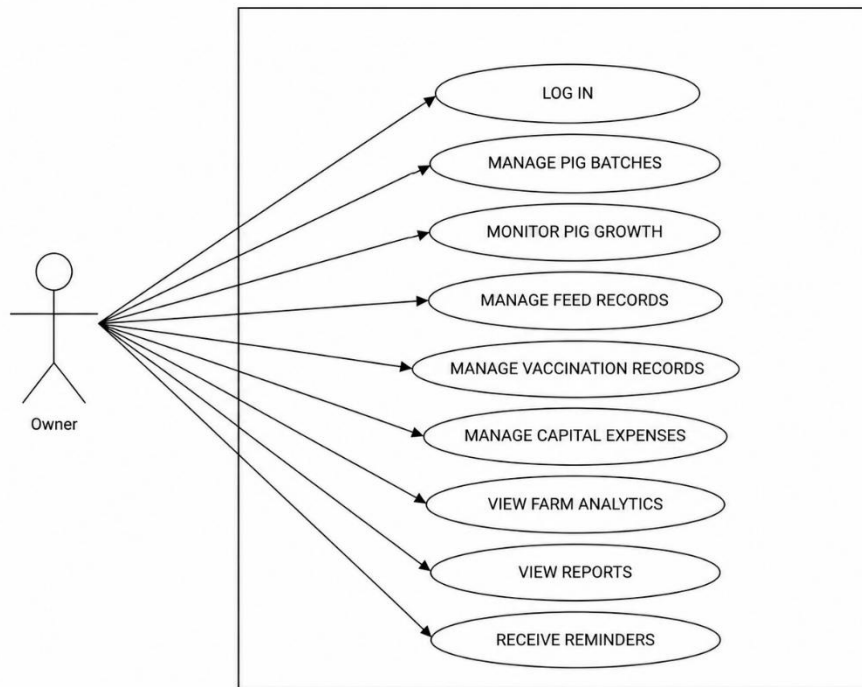


Figure 4 illustrates the Use Case Diagram for the PrepAPig System. The diagram shows the interactions between the Owner, who serves as the primary actor, and the different system functionalities designed to support pig growth monitoring, feed management, vaccination tracking, expense monitoring, and farm analytics. The system aims to improve farm management efficiency by digitizing farm records, organizing monitoring activities, and providing analytical reports to support decision-making.

The Log In use case allows the owner to securely access the system before performing any monitoring and management activities. This ensures that farm records and

information are only accessible to authorized users. After successfully accessing the system, the owner can proceed to the Manage Pig Batches use case, which enables the recording and organization of pig batch information such as batch name, pig quantity, age, and monitoring details. This functionality ensures that pig records are systematically maintained within the system.

The Monitor Pig Growth use case enables the owner to observe pig development using recorded age and weight information. Through this functionality, the owner can monitor growth progress and identify trends that may affect feeding and farm management decisions. The Manage Feed Records use case allows the owner to record and monitor feed consumption per batch, helping maintain organized feeding records and supporting feed forecasting activities. Similarly, the Manage Vaccination Records use case enables the owner to manage vaccination schedules, monitor vaccination status, and maintain vaccination history records for each pig batch.

The Manage Capital Expenses use case supports the recording and monitoring of farm-related expenses such as feeds and vaccines. This functionality assists the owner in tracking operational costs and maintaining organized financial records. The View Farm Analytics use case allows the owner to analyze farm data such as growth trends, feed consumption patterns, and forecasting summaries generated from collected records. In addition, the View Reports use case enables the generation and retrieval of reports related to pig growth, feed consumption, vaccination activities, and overall farm performance.

Lastly, the Receive Reminders use case allows the system to notify the owner regarding upcoming vaccinations, feed schedule changes, and overdue farm activities to reduce the risk of missed tasks and improve farm management efficiency.

3.2.5 Non-Functional Requirements

1. PERFORMANCE

- 1.1 The system will load the pig batch records dashboard within 3 seconds under normal network conditions on smartphones, tablets, and computers.
- 1.2 The system will process and display growth trend analytics from recorded weight and age data for a single batch within 5 seconds.
- 1.3 The system will generate feed schedules and reminders for up to 10 pig batches simultaneously without noticeable delay.
- 1.4 The system will synchronize offline recorded data to the cloud within 30 seconds after internet connectivity is restored.

2. SECURITY

- 2.1 The system will require secure login using a username and password for owner account before accessing any farm records.
- 2.2 The system will encrypt all data transmitted between the client device and the server using HTTPS protocol.
- 2.3 The system will prevent unauthorized access to vaccination, feed, and expense records by ensuring that only authenticated owners can view or modify data.

3. USABILITY

- 3.1 The system will provide a simple, farm-friendly interface with font sizes and buttons that are easily readable and tappable on mobile devices (minimum touch target size of 44x44 pixels).
- 3.2 The system will include confirmation dialogs before deleting any pig batch, vaccination record, or expense entry to prevent accidental data loss.

3.3 The system will display clear error messages when offline data synchronization fails, including instructions for the owner to retry.

3.4 The system will support offline access to recently viewed pig growth data, feed records, and vaccination schedules without requiring an internet connection.

4. RELIABILITY

4.1 The system will maintain 99% uptime during daily farm operating hours (6:00 AM to 8:00 PM) when deployed for actual use.

4.2 The system will automatically save owner input data locally on the device if the internet connection is lost while recording feed consumption or vaccination status.

4.3 The system will not lose recorded data during unexpected application closure; any unsynced entries will be preserved and retried upon reopening.

5. AVAILABILITY

5.1 The system will be accessible via Progressive Web App (PWA) on smartphones, tablets, and computers without requiring app store installation.

5.2 The system will allow the owner to access pig batch records and generate reports even when the server is temporarily unreachable, using cached data.

6. MAINTAINABILITY

6.1 The system will log any failed data synchronization attempts and reminder generation errors for troubleshooting by the development team.

6.2 The system will allow future addition of new analytics features (e.g., disease outbreak prediction) without modifying the existing pig growth monitoring module.

7. ACCURACY

7.1 The system will calculate feed consumption summaries and feed forecasting with no rounding errors that would affect farm expenditure decisions.

7.2 The system will correctly compute pig age in days and weeks based on the recorded monitoring dates, with deviation not exceeding one day.

7.3 The system will generate reminders for upcoming vaccinations based on the exact schedule entered by the owner, accurate to the specified date.

3.2.6 System Context

This subsection presents the overall interaction between the PrepAPig system and the external entities that provide inputs to and receive outputs from the system. The context diagram defines the boundary of PrepAPig as an AI-powered pig growth tracking and farm management system. It identifies the users and external services that interact with the system and shows how pig growth records, feed monitoring, vaccination schedules, and analytics are managed within the system.

Figure 5.
Context Diagram

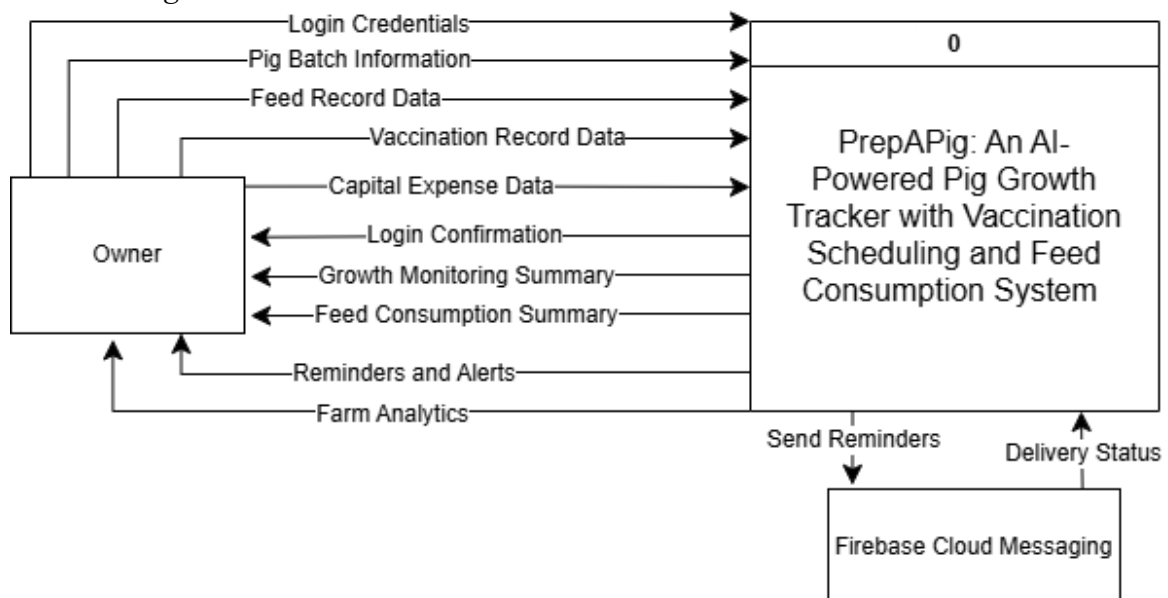


Figure 5 illustrates the Context Diagram for the PrepAPig system. The diagram shows the high-level interaction between the Owner, the primary external entity, along with the secondary external notification service (Firebase Cloud Messaging), and the system. It identifies all major data flows entering and leaving the system boundary, reflecting the core functional requirements defined for pig growth monitoring, feed consumption tracking, vaccination scheduling, and farm analytics reporting.

On the left side of the diagram, the Owner initiates several inputs to the PrepAPig system. These inputs include Login Credentials, which are necessary for authentication and secure access to the system. The Owner also provides Pig Batch Information, containing details such as batch name, number of pigs, age, weight, and monitoring dates to support growth tracking. Additionally, Feed Record Data is supplied to document feed consumption per batch, while Vaccination Record Data captures vaccination schedules and completion status. Finally, Capital Expense Data is entered to record costs incurred for feeds, vaccines, and piglets, enabling basic financial analysis.

On the right side of the diagram, the PrepAPig system provides several outputs back to the Owner. The system generates a Login Confirmation, indicating successful or failed authentication. It produces a Growth Monitoring Summary, which compiles recorded weight progress and developmental trends per batch, supporting the owner's ability to observe performance without manual computation. The Feed Consumption Summary delivers organized data on feed usage patterns over time, aiding in cost management and inventory planning. The system also issues Reminders and Alerts for upcoming vaccinations, feeding schedules, and overdue tasks, improving task organization and reducing missed schedules. Furthermore, the system supplies Farm Analytics, which

include growth trend analyses, feed consumption pattern recognition, feed forecasts, and profit assessments. These outputs are essential for data-driven decision-making, operational efficiency, and long-term farm profitability.

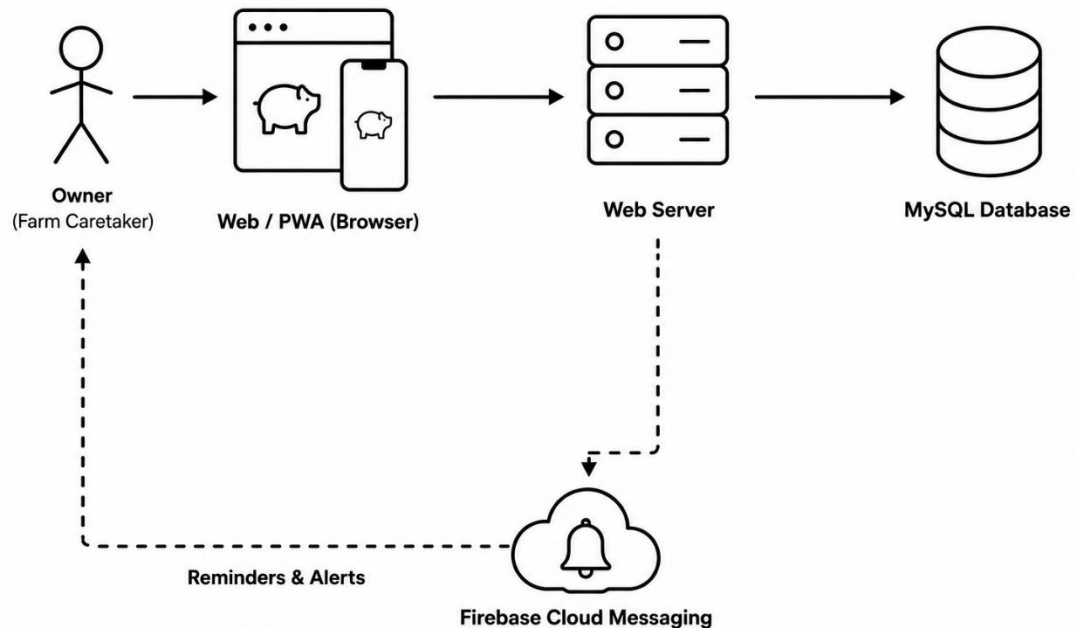
The diagram also includes the interaction between the PrepAPig system and the Firebase Cloud Messaging external notification service. The system transmits Send Reminders containing notification payloads and target device tokens to this service. In return, Firebase Cloud Messaging returns Delivery Status messages to the system, confirming successful transmission or reporting errors for each reminder. This flow ensures that critical farm notifications are reliably delivered to the Owner's device without requiring the application to remain open.

Overall, the Context Diagram presents a clear and simplified view of the system's external interactions. It demonstrates how the Owner relies on the PrepAPig system to automate pig growth tracking, feed consumption monitoring, vaccination scheduling, and analytics reporting, while also leveraging external notification infrastructure to deliver timely reminders. The diagram effectively defines the system boundary, includes all relevant external entities, and explicitly indicates all inputs and outputs, supporting informed and data-driven livestock management.

3.3 System Architecture

This section presents the overall structure of the proposed system, including system components, data flow, and user roles.

Figure 6.
System Architecture Design



As shown in Figure 6, the System Architecture of PrepAPig consists of multiple layers that interface seamlessly to support farm management operations. At the user level, the Owner (Farm Caretaker) interacts directly with the Web or Progressive Web Application (PWA) running on a browser, which serves as the primary interface for recording pig batches, monitoring growth, logging feed consumption, scheduling vaccinations, tracking expenses, and generating reports. At the core of the system, the Web Server acts as the central hub, receiving requests from the browser, executing application logic for pig growth monitoring, feed management, and vaccination scheduling, and coordinating with the Reminders & Alerts module. The Reminders & Alerts component evaluates scheduled tasks and overdue activities, then triggers notifications through Firebase Cloud Messaging (FCM), which delivers push notifications to the Owner's

device. The MySQL Database plays a significant role in storing and organizing all critical information, including pig batch records, growth history, feed logs, vaccination schedules, expense entries, and user credentials, ensuring data persistence and consistency for the entire system.

3.3.3 Data Flow

Data within PrepAPig flows through four primary operational pathways, each corresponding to a distinct farm management task or system function.

In the farm record management pathway, the farm owner accesses the PrepAPig Progressive Web Application through a web browser. The application sends the entered pig batch information, growth weights, feed consumption records, vaccination schedules, and capital expenses to the Node.js backend, which validates the data and writes it into the appropriate MySQL tables—pig_batches, growth_records, feed_logs, vaccination_schedules, and expenses. A confirmation is returned to the interface, and the updated records immediately appear on the owner's dashboard, ensuring that all farm information is stored centrally and consistently.

In the analytics generation pathway, the owner requests a growth trend chart, feed consumption summary, or feed forecast from the analytics dashboard. The frontend sends the request to the backend's analytics endpoint, which retrieves the relevant historical weight and feed records from the MySQL database. The integrated TensorFlow.js model then computes the growth trend line and feed forecast based on the retrieved data. The results are sent back to the frontend as structured data and rendered as interactive charts using Chart.js, providing the owner with clear, data-driven insights without modifying any stored records.

In the notification handling pathway, a server-side scheduled process automatically checks the MySQL database for upcoming vaccination dates, feed schedule changes, and overdue tasks. When a reminder condition is met, the backend constructs a notification payload containing the relevant task details and delivers it to the owner's mobile device or desktop browser through Firebase Cloud Messaging. The same reminders also appear as in-app alerts on the dashboard, so that critical farm activities are never missed.

In the user authentication pathway, the owner submits login credentials through the application's login form. The backend verifies the username and password against the stored user account in the MySQL database. Upon successful authentication, a session token is issued, granting seamless access to all protected modules—pig management, feed and vaccination tracking, expense monitoring, and analytics—for the duration of the session. If the session remains idle for an extended period, it expires automatically, and the owner must re-authenticate to continue using the system.

System Components and Modules System Modules and Components

1. Pig Management Interface

This is a private module accessible only after the farm owner logs in. It allows the owner to add, update, and view pig batch information, including batch name, number of pigs, age, and weight. The interface records growth weights over time and supports offline data entry through the PWA service worker. Batch management includes creating new pig batches with unique identifiers, start dates, initial weights, and batch counts. Growth tracking allows periodic weight measurements for individual pigs or batches, with historical weight progression and average daily gain calculation. Offline weight entry ensures that when internet connectivity is unavailable, the owner can still record weight

data; the module stores entries locally using IndexedDB and synchronizes them automatically once the connection is restored.

2. Feed and Vaccination Manager

This is a private module accessible only after login. It allows the owner to manage daily feed consumption records, schedule vaccination dates, mark completed vaccinations, and receive automated reminders. Key functions include:

2.1 Feed Logging – Record daily feed intake per pig batch, track cumulative consumption, and view historical feed usage patterns to support forecasting.

2.2 Vaccination Scheduling – Set vaccination due dates, record vaccine types and doses, mark administration status, and receive push notifications for upcoming or overdue vaccinations through Firebase Cloud Messaging.

2.3 Automated Reminders – A backend cron job checks the database each day for upcoming vaccinations (within two days) or overdue tasks. The system then sends a push notification to the owner's registered device.

3. Expense Tracker

This is a private module accessible only after login. It allows the farm owner to record capital expenses including feed costs, vaccine purchases, and piglet acquisitions. The module integrates with the analytics dashboard to compute cost per pig, feeding efficiency, and overall farm profitability. Key functions include:

3.1 Expense Entry – Record expenses by category (feeds, vaccines, piglets, miscellaneous), including amount, date, and associated pig batch.

3.2 Expense Summary – View total expenses per batch, per time period, or per category.

3.3 Profit Calculation – The system combines expense data with growth performance to calculate profit margins per batch and overall farm return on investment.

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4. MySQL Database

This is the internal data storage layer of PrepAPig. It is a relational database management system that stores all persistent farm data. The database includes tables for pig batches (batch name, count, age, monitoring dates), growth records (weight measurements per batch over time), feed consumption logs (daily intake per batch), vaccination schedules (vaccine type, due date, completion status), capital expenses (costs of feeds, vaccines, and piglets), and user credentials (hashed password and authentication token). The database enforces referential integrity and ACID compliance to ensure consistent records across all modules. It is accessed internally by the Node.js backend through the mysql2 connection pool and is never exposed directly to the frontend.

5. Authentication Service

This internal component handles secure login for the farm owner. It is implemented using JSON Web Tokens (JWT) within the Node.js backend. When the owner submits login credentials, the service validates the username and password against the user table in the MySQL database. Upon successful validation, it generates a signed JWT that encodes the owner's role and session information. The token is returned to the frontend and must be included in the Authorization header of all subsequent API requests to private modules. Tokens expire after a defined inactivity period, requiring re-authentication for continued access.

6. Notification Service (FCM)

This internal component integrates Firebase Cloud Messaging to deliver push notifications to the farm owner's mobile device or desktop browser. It operates in the Node.js backend and is triggered by a scheduled cron job that runs three times daily (morning, midday, evening). The job queries the MySQL database for upcoming vaccination dates (within two days), feed schedule changes, and overdue tasks. For each eligible reminder, the service constructs a notification payload containing a title, body, and associated data (e.g., vaccination due for Batch A). It then calls the Firebase Admin SDK to send the payload to the owner's stored device token. The service also logs delivery status and retries failed notifications.

7. AI Integration Layer

This internal component provides the artificial intelligence features of PrepAPig. It is embedded within the Node.js backend and uses TensorFlow.js for machine learning inference, along with json-rules-engine for rule-based alerts. Key functions include:

7.1 Growth Trend Analysis – When the owner requests a growth chart for a pig batch, the module retrieves historical weight-vs-age data from the MySQL database. It then creates a linear regression model using TensorFlow.js (with a single dense layer trained for 100 epochs on the fly) and computes the slope and intercept of the weight trend line. The result is returned to the frontend for visualization using Chart.js.

7.2 Feed Forecasting – Similarly, the module applies a linear regression model to historical feed consumption records to predict total feed required for the next seven days. The forecast is displayed on the analytics dashboard.

7.3 Rule-Based Alerts – The module uses json-rules-engine with a static set of pre-defined rules (e.g., “if average daily gain is below target, send alert”). It evaluates these rules against the computed trend outputs and generates textual insights that appear on the dashboard.

8. Offline Synchronization Module

This internal client-side component is implemented through the PWA service worker and IndexedDB. When the farm owner records data while offline, for example entering a new weight measurement or feed log without internet, the module intercepts the write operation and stores it locally in IndexedDB as a pending queue item. The service worker registers a background sync event. Once internet connectivity is restored, the background sync triggers and the module replays each queued operation to the server via a POST request to the /api/sync endpoint. The server processes each operation (e.g., inserting a growth record) and returns success or conflict. The module retries failed operations and cleans the queue upon success. This design ensures zero data loss even when the farm has unreliable internet access.

User Roles

PrepAPig supports a single user role, the Owner (Farm Caretaker), with full access to all system functions enforced through JWT-based authentication and row-level access controls within the MySQL database. The Owner must log in using valid credentials before accessing any private module, including the pig management interface, feed and vaccination manager, expense tracker, analytics dashboard, and offline synchronization features. No public or unauthenticated access is permitted for any data entry or viewing function. Each successful login generates a signed JSON Web Token that is required for

all subsequent API requests; the token expires after a defined period of inactivity, requiring re-authentication for continued access.

The Owner holds complete system access, including the ability to create, read, update, and delete all farm records. This includes adding and managing pig batch information (batch name, number of pigs, age, weight, monitoring dates), recording daily feed consumption per batch, scheduling vaccination due dates and marking completed vaccinations, entering capital expenses (feeds, vaccines, piglet costs), viewing AI-generated growth trend analysis and feed forecasts, generating performance reports, and receiving push notifications for upcoming or overdue tasks. The Owner can also access the offline synchronization module, which stores pending write operations locally when internet connectivity is unavailable and replays them automatically once the connection is restored.

Because the system is designed exclusively for a single farm operator at ANCY's Pig Farm, there is no distinction between administrator, agent, or visitor roles. The Owner has no visibility into any other user's data because no other user accounts exist. All data stored in the MySQL database, including pig batches, growth records, feed logs, vaccination schedules, expenses, and user credentials, is owned and fully accessible only to the authenticated Owner. This design ensures data privacy and eliminates the complexity of multi-user permission management while still providing complete farm management functionality within a secure, offline-capable progressive web application.

Figure 7
Data Flow Diagram

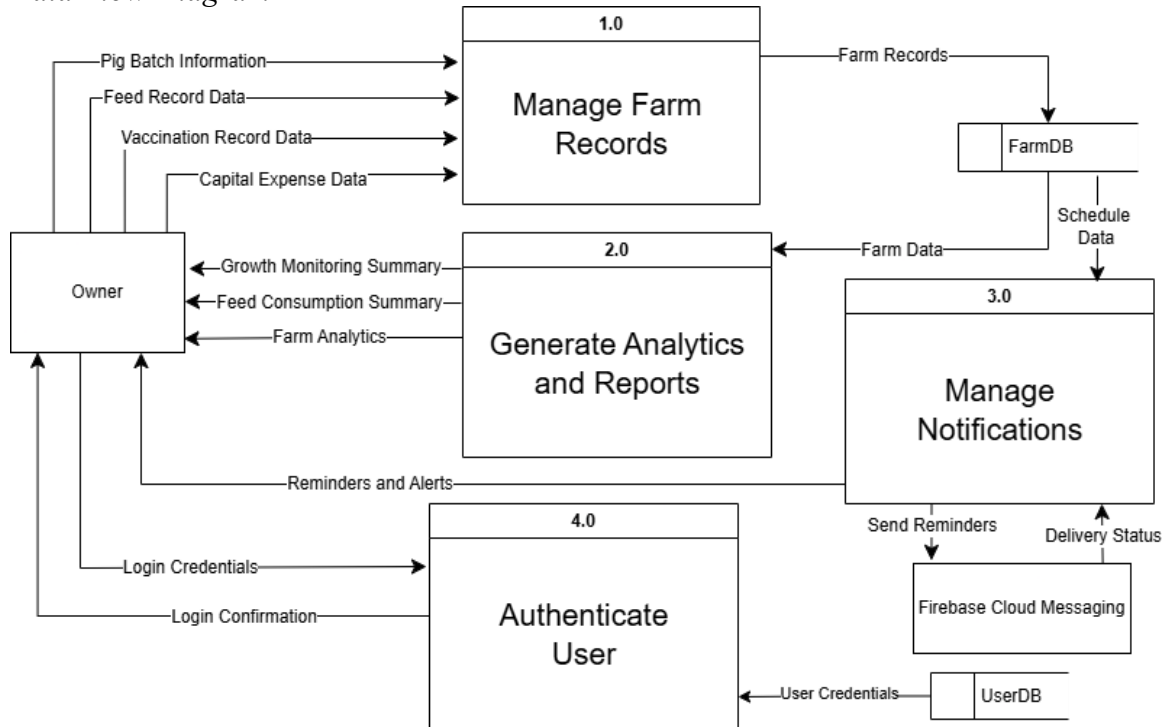


Figure 7 illustrates the data flow of the PrepAPig system for ANCY’s Pig Farm. It describes the system’s processing components, the centralized MySQL database storage, and the actions taken by the external actor, the Owner (Farm Caretaker), as well as the external Firebase Cloud Messaging service. The different procedures involved are listed below.

Process 1: Manage Farm Records – The Owner sends pig batch information, feed record data, vaccination record data, and capital expense data to the platform. The system validates the input and writes these records to the MySQL Database (D1). No data is returned to the Owner during this write operation, but a confirmation message is displayed on the interface. This process ensures that all farm records are stored centrally and consistently.

Process 2: Generate Analytics and Reports – When the Owner requests a growth summary, feed summary, or farm analytics, the system retrieves the relevant historical data from D1, including pig weights, feed logs, and expense records. The built-in AI layer (TensorFlow.js) computes growth trends and feed forecasts using linear regression models. The system then returns the processed analytics to the Owner as visualized charts and summary tables. This process does not modify any stored data; it only reads from D1 and displays outputs.

Process 3: Manage Notifications – The system runs a scheduled cron job that periodically reads schedule data from D1, specifically upcoming vaccination dates, feed schedule changes, and overdue tasks. For each eligible reminder, the system composes a notification payload and sends it to the Firebase Cloud Messaging service (D2). FCM returns a delivery status (success or failure) back to the system. The system also displays reminders and alerts directly to the Owner within the PWA interface. This process ensures that critical farm tasks are not missed.

Process 4: Authenticate User – The Owner submits login credentials to the system. The system reads the stored user credentials from D1 and verifies the submitted username and password. If the credentials match, the system generates a signed JSON Web Token and returns a login confirmation to the Owner. If the credentials do not match, an error message is returned. This process secures all private modules and ensures that only the authorized farm owner can access sensitive farm data.

3.4 AI Integration Design

The AI integration layer of PrepAPig comprises three interconnected components: a growth trend analysis engine, a feed consumption forecasting module, and a rule-based

descriptive analytics engine. Each component is designed to operate within the system's web-based architecture using pre-built JavaScript libraries (TensorFlow.js and json-rules-engine), eliminating the need for custom model training or dedicated GPU infrastructure while maintaining acceptable performance for an academic capstone project. This section describes the design, data inputs, processing logic, and output integration of each AI component.

Growth Trend Analysis Engine

The growth trend analysis engine generates a visual representation of pig weight progression over time for each batch or individual pig. The engine analyzes historical records stored in the MySQL database, specifically the pig weights recorded at different ages, the batch identifier, and the date of each weight entry. Using the pre-built TensorFlow.js library (without any custom training), the engine produces a trend line that shows whether growth is progressing as expected, below target, or above target. For batches with very few weight records, a simple message informs the user that more data is needed. The resulting trend line is sent to the React frontend via an API and displayed on the pig monitoring dashboard using Chart.js.

Feed Consumption Forecasting Module

The feed consumption forecasting module predicts the amount of feed required for a given batch over a defined future period. The module examines historical feed consumption records, including daily or weekly feed amounts, pig age, and batch size. Using the same pre-built TensorFlow.js library (again without any custom training), it produces a forecasted quantity (expressed as a total kilogram amount for the upcoming interval). If historical feed records are sparse, the module shows a simple average of recent

consumption values as a conservative estimate. The forecast is displayed on the feed management dashboard, helping farm owners plan feed purchases and adjust inventory.

Rule-Based Descriptive Analytics Engine

The rule-based descriptive analytics engine generates plain-language insights, alerts, and reminders based on the outputs of the growth trend engine, the feed forecasting module, and directly from farm records. The engine uses the json-rules-engine library, which applies a fixed set of conditional rules defined before system deployment. Example outputs include: “Growth is slower than expected. Consider checking feed quality or health” when the trend line indicates slower-than-normal growth; and a reminder notification when a vaccination is due within three days. The engine runs after each growth or forecast calculation, as well as on a daily schedule to check for upcoming tasks. Generated insights and alerts are pushed to the React frontend via an API and displayed on the dashboard’s insights panel; vaccination reminders are also sent as push notifications through Firebase Cloud Messaging.

All three AI components rely entirely on pre-built libraries (TensorFlow.js and json-rules-engine) and require no custom model training, hyperparameter tuning, or specialized hardware. This design ensures that the AI integration remains feasible within the scope of a Bachelor of Science in Information Technology capstone project while providing meaningful decision support for farm owners.

3.5 Development Tools and Technologies

This section lists and describes the tools used in system development. Every technology is chosen considering the functional requirements of the system such as being

offline capable, having data consistency, being able to give real-time alerts, and cross-platform compatibility.

Programming Languages

The system will be developed using JavaScript for both frontend and backend. JavaScript enables a unified language across the entire stack, simplifying development and maintenance. Its non-blocking, event-driven model will be well-suited for handling farm data entry, real-time reminders, and offline synchronisation without performance bottlenecks. By using JavaScript on both the client and server sides, the development team will reduce context switching and streamline code reuse. This unified approach also ensures that features such as data validation, API calls, and background synchronisation behave consistently regardless of where they execute.

Frameworks

The frontend will be implemented with React. React's virtual DOM and component-based architecture enable efficient, scalable user interfaces (Komperla et al., 2021). For PrepAPig, this will mean reusable components for growth dashboards, feeding logs, and vaccination schedules, all updating smoothly even on mobile devices with limited processing power. Each component (such as a pig weight entry form or a feed consumption chart) will be developed in isolation and reused across different parts of the application. This modularity will make the code easier to test, maintain, and extend as farm management needs evolve. The virtual DOM will minimise direct manipulation of the browser's DOM, reducing page reloads and providing a near-native user experience.

Express.js will be used as the backend web framework. It is lightweight and flexible, managing several simultaneous requests efficiently. PrepAPig will leverage

Express.js to create RESTful API routes for pig batch management, feed consumption records, vaccinations, expenses, and reporting, while also handling authentication middleware and error handling. Each API endpoint will be structured to accept and return JSON data, making integration with the React frontend straightforward. Express.js middleware will verify user permissions before processing requests, ensuring that only authorised farm staff can access sensitive data.

Tailwind CSS will provide a utility-first approach to styling. It gives complete control over customisation without pre-built components, and its PurgeCSS tool will remove unused styles. For PrepAPig, this will ensure fast loading on low-end devices and a responsive layout that works on desktops, tablets, and phones. Developers will apply utility classes directly in the JSX markup, eliminating the need to switch between separate CSS files. The resulting style bundle will be very small because unused classes are automatically stripped during the build process. This lightweight approach contributes to the overall offline-first strategy by reducing the amount of data that must be cached.

Frontend and Backend Technologies

At its core, PrepAPig will use HTML5. Its native support for offline caching, form validation, and the <canvas> element makes it suitable for farm management interfaces that must work without constant internet and present clear data visualisations. The offline application cache will allow the system to function during network interruptions, ensuring that farm caretakers can still access essential records even when connectivity is lost. Built-in form validation will ensure accurate entry of pig weights, vaccination dates, and feed amounts, reducing data entry errors. Semantic tags such as <header>, <nav>, <section>, and <footer> will help organise the user interface into logical sections,

improving navigation and overall usability across desktop and mobile devices.

PrepAPig will be built as a Progressive Web Application (PWA). PWAs use service workers and web app manifests to provide offline operation, push notifications, and background sync (Turansky & Leshchuk, 2020). This will make the system ideal for farm environments with unstable internet so that the owner will be able to record pig data offline, receive vaccination alerts via push notifications, and automatically synchronise when connectivity resumes. The service worker will intercept network requests and serve cached responses when offline, ensuring that the dashboard and forms remain usable. The web app manifest will allow farm owners to install PrepAPig on their device's home screen, providing quick, app-like access without visiting an app store.

Database

MySQL was selected as the database because it is lightweight, open-source, runs on low hardware specifications, and offers strong data protection. It will store all farm-related data (pig batches, feed logs, vaccinations, expenses, user accounts). Its ACID compliance will keep financial and health records consistent, even in the event of power outages or system crashes. Complex SQL queries will be used to generate reports such as monthly feed consumption trends, batch growth comparisons, and profit margin calculations. Because MySQL supports concurrent multi-user access, several farm workers can update data simultaneously without corrupting the database.

Vercel will be the deployment platform. It offers GitHub-integrated CI/CD, a global CDN, automatic scalability, and built-in PWA support. These features will ensure fast load times regardless of the user's location, automatic updates on every code push, and seamless handling of increased traffic during peak farm management periods. Every time the

development team pushes code to the GitHub repository, Vercel will automatically build and deploy the new version, eliminating manual deployment steps. The global CDN will cache static assets (HTML, CSS, JavaScript) at edge locations worldwide, so farm owners in remote areas will still experience quick loading. Automatic scaling will handle traffic spikes, such as when many farmers log in to record weights before a market deadline.

Chart.js will be used for data visualisation. It is an open-source library that renders charts using HTML5 Canvas with a simple configuration. PrepAPig will use it to display growth trends, feed consumption graphs, and profit distributions, making farm data easy to understand at a glance. Farm owners will be able to view a line chart showing weight progression over weeks, a bar chart comparing feed consumption across batches, or a pie chart illustrating expense categories. The library's built-in tooltips and animations will make the dashboards more intuitive without requiring complex custom coding. Because Chart.js renders directly to the Canvas element, performance will remain smooth even on low-powered devices.

API's including AI tools

PrepAPig will integrate TensorFlow.js for growth trend analysis and feed forecasting. The library will apply pre-defined linear regression models directly in the Node.js backend, eliminating the need for separate Python-based ML services (Pires, 2023). This will keep the entire system within the JavaScript ecosystem, reduce infrastructure complexity, and provide reliable forecasts based on historical pig weight and feed consumption data – all without custom model training. The growth trend analysis will help farmers identify pigs that are not gaining weight as expected, enabling early intervention. The feed forecasting feature will estimate future feed requirements, helping

farm owners optimise purchasing and reduce waste. Because TensorFlow.js runs in the same Node.js process as the rest of the backend, there will be no network latency or additional cost for specialised ML servers.

Firebase Cloud Messaging serves as to send push notifications. FCM reliably delivers messages over HTTPS even when the browser is closed or the application is inactive (Jiddere et al., 2025). This will guarantee that the owner (farm caretaker) receives timely reminders for vaccinations and feeding schedules, reducing missed tasks. Notifications will be triggered by the backend when a vaccination date approaches or when a feeding schedule is overdue. Because FCM works across Android, iOS, and web platforms, alerts will be received regardless of the device they use. This cross-platform support is essential for farms where workers may have different types of mobile phones.

3.6 Implementation Procedure

This section presents the step-by-step process of system development, including the system development stages and the integration process. Development is organized into six iterative Sprints, each delivering a working increment that is reviewed by the farm owner before the next Sprint begins.

Sprint 1 – Environment Setup and Authentication Infrastructure

Sprint 1 covers the project environment setup and authentication infrastructure. The team initializes the Node.js and React project with Tailwind CSS, creates the MySQL database schema for pig batches, growth records, feed logs, vaccination schedules, and expenses, and establishes the JWT-based authentication system for the farm owner. The Progressive Web App shell is built with responsive layout and basic offline caching. This

sprint produces the foundational infrastructure on which all subsequent modules are built, and the owner can securely log in to a functional application.

Sprint 2 – Pig Batch Management and Growth Monitoring

Sprint 2 covers the pig batch management and growth monitoring module. The team builds the interface for creating and organizing pig batches, implements weight recording over time, and develops the growth history dashboard. All pig and weight data is stored centrally in MySQL and displayed immediately after entry. This sprint delivers a fully functional digital pig batch and growth tracking flow, replacing the farm's handwritten logs for this activity.

Sprint 3 – Feed and Vaccination Management

Sprint 3 covers the feed consumption logging and vaccination scheduling modules. The team develops the feed entry form that records daily feed amounts per batch and tracks cumulative consumption, and builds the vaccination scheduling interface where the owner can set dates, record vaccine types, and mark completed vaccinations. Both modules are connected to the MySQL backend with input validation. This sprint produces a structured feed and vaccination management flow that mirrors the farm's actual daily routines.

Sprint 4 – Expense Tracking and Basic Reports

Sprint 4 covers the expense tracking module and basic operational reports. The team builds the expense entry interface for logging costs of feeds, vaccines, and piglets, and creates summary views that display total expenses by batch and by category. Basic reports on feed consumption and vaccination completion are generated from the stored data. This sprint provides the owner with clear financial visibility and report outputs for the first time.

Sprint 5 – AI-Powered Analytics and Reminders

Sprint 5 covers the integration of AI-assisted features and automated reminders. The team integrates TensorFlow.js to process historical weight and feed records and generate growth trend charts and feed requirement forecasts. A rule-based alert engine is configured to flag conditions such as slower than expected growth. Firebase Cloud Messaging is set up to push vaccination and feeding reminders directly to the owner's device. This sprint delivers AI-powered insights and automated notifications that support data-driven farm decisions.

Sprint 6 – Offline Reliability, Notification Tuning, and Deployment Preparation

Sprint 6 covers offline data collection, notification refinement, and system deployment. The team strengthens the offline capability so the owner can enter weight, feed, and vaccination records without internet, with automatic synchronization upon reconnection. Notification delivery is fine-tuned for reliability, and the user interface is polished with confirmation dialogs and clear feedback messages. The system is deployed to a cloud platform, backend host, and cloud database. A hands-on orientation session ensures the farm owner can confidently use the system. This sprint produces the final, polished PrepAPig system ready for daily farm operations.

The integration of system components is performed incrementally alongside the Sprints. The frontend and backend are connected through secure routes that handle all farm data operations, with the database acting as the central store for pig records, growth logs, feed data, vaccination schedules, and expenses. The AI analytics module is linked to the same database, reading historical information to generate trends and forecasts on demand, and the results are displayed on the dashboard without modifying the stored data. The

notification service is integrated to automatically monitor the database for upcoming and overdue schedules, delivering reminders through the cloud messaging platform. Offline data capture is woven into the frontend so that entries are saved locally when the network is unavailable and synchronized with the server once connectivity is restored. Each integration point is verified by executing complete farm workflows—logging in, recording data with and without internet, receiving reminders, and viewing updated charts—to confirm that all components work together seamlessly.

3.7 Testing and Evaluation

This section describes the testing and evaluation procedures applied to the PrepAPig system. The testing process follows the Agile Development Methodology described in Section 3.1, wherein testing occurs continuously throughout each iteration rather than as a single phase at the end. The evaluation incorporates multiple testing methods and metrics to assess the system from technical and user perspectives.

Functional Testing

Functional testing verifies that each system feature operates according to the functional requirements specified in Section 3.2.4. Test cases are written for core functionalities including pig batch management, growth weight recording, feed consumption logging, vaccination scheduling, expense tracking, and report generation. Each test case defines expected inputs, execution steps, and expected outputs. Baseline data from the farm owner's existing handwritten records simulate real farm scenarios. Any discrepancy between actual and expected behavior is documented and addressed in the subsequent development iteration. Functional testing also covers the AI-assisted features,

specifically growth trend analysis and feed forecasting, by comparing the system's computed trends against manually calculated values derived from historical records.

Usability Testing

Usability testing assesses the ease with which the farm owner and caretakers navigate the Progressive Web App, enter data, view dashboards, and receive reminders. Participants from ANCY's Pig Farm, comprising the farm owner and one caretaker, perform typical daily tasks such as adding a new pig batch, recording a feed consumption entry, checking vaccination schedules, and viewing a growth chart. Observers record task completion time, number of errors encountered, and points of confusion. Following the tasks, participants complete a questionnaire based on the System Usability Scale (SUS) to produce a quantitative usability score. The system must achieve a SUS score of at least 70 to be considered acceptable for farm use. Feedback from this test guides improvements to button sizes, form layouts, error messages, and the overall interface.

User Acceptance Testing (UAT)

User acceptance testing occurs after all core features have been implemented and functionally tested. The farm owner uses the PrepAPig system in the actual farm environment for two weeks to manage pig batches, feed logs, vaccination schedules, and farm expenses. During this period, the owner receives push notifications for upcoming vaccinations and feeding tasks. At the end of the two-week trial, a structured interview and feedback session determines whether the system meets the owner's expectations and improves farm management efficiency compared to the existing manual method. Acceptance criteria include the ability to record data without technical assistance, reliable

notification delivery, and confidence in report accuracy. The farm owner signs a user acceptance sign-off document upon successful completion of UAT.

Performance Testing

Performance testing evaluates the system's responsiveness, throughput, and resource usage under normal and peak conditions. Although the system is designed for a single user, performance testing remains essential to ensure a smooth experience. The following performance metrics are measured and validated against the non-functional requirements in Section 3.2.5:

Response time – The time to load the pig batch records dashboard under normal network conditions (home Wi-Fi or 4G) must not exceed three seconds. The time to process and display growth trend analytics for a single batch must not exceed five seconds. These times are measured using browser developer tools (Network tab) and repeated ten times to calculate an average.

System performance – The system is tested while generating feed schedules and reminders for up to ten pig batches simultaneously. Server CPU and memory usage are monitored using Node.js built-in process methods. The time to synchronize offline recorded data after restoring internet connectivity must not exceed thirty seconds, as required by the non-functional requirements.

Accuracy or correctness – Feed consumption summaries and feed forecasting calculations are validated against manual computations using the same source data. Pig age calculation in days and weeks must not deviate by more than one day from the actual calendar difference. Reminders for upcoming vaccinations are triggered exactly on the

scheduled date as entered by the owner. These accuracy checks are performed using automated unit tests and manual verification with baseline farm records.

User Satisfaction

User satisfaction is measured as a separate evaluation metric through a post-study questionnaire administered after the UAT period. The questionnaire includes statements such as “The system helps me remember vaccination schedules,” “I find the growth charts easy to understand,” and “I would recommend this system to other pig farmers.” Responses are collected using a five-point Likert scale ranging from “Strongly Disagree” to “Strongly Agree.” The average score across all questions is calculated, with a target of at least 4.0 (Agree) to indicate satisfactory user satisfaction.

Adviser Monitoring Requirement

Progress is reviewed periodically by the research adviser throughout the testing and evaluation process. Before proceeding from one testing phase to the next (for example, from functional testing to usability testing, and from usability testing to user acceptance testing), the adviser must validate the outputs. Validation includes reviewing test cases and results, observing usability testing sessions when possible, examining the UAT sign-off document, and verifying that performance metrics meet the specified thresholds. The next phase commences only after receiving documented approval from the adviser. This monitoring ensures rigorous testing and confirms that the system is ready for actual deployment to ANCY's Pig Farm.

All testing results are compiled into a testing report that includes test case logs, performance measurement tables, usability scores, UAT feedback, and adviser validation signatures. Any identified issues that cannot be resolved before deployment are

documented as known limitations and addressed in the recommendations for future improvements.

3.8 Evaluation Methods

This section explains how the data collected from the testing and evaluation procedures described in Section 3.7 will be analyzed and interpreted. The evaluation of PrepAPig employs three complementary analytical methods, namely the Likert scale, weighted mean, and descriptive analysis, applied in combination to produce a comprehensive and structured assessment of the system's quality from the perspective of the farm owner and caretaker.

Likert Scale

The primary instrument used to collect evaluation data from the farm owner after user acceptance testing is a four-point Likert scale. Each item in the evaluation questionnaire presents a statement about a specific aspect of PrepAPig's quality, such as the ease of recording pig batch information, the reliability of vaccination reminders, the clarity of growth trend charts, the accuracy of feed forecasting, or the usefulness of offline data entry. The respondent indicates their level of agreement using one of four response options: 4 for Strongly Agree, 3 for Agree, 2 for Disagree, and 1 for Strongly Disagree. The four-point scale removes the neutral option, which encourages the farm owner to express a definite opinion rather than selecting a middle response. The Likert scale is selected as the evaluation instrument because it provides a standardized, quantifiable measure of user perception that can be aggregated and compared across different functional areas of the system. The questionnaire items are organized according to the key functional modules of PrepAPig, including pig batch management, feed consumption logging,

vaccination scheduling, expense tracking, AI-based growth trend analysis and feed forecasting, offline synchronization, and push notification delivery. This ensures that the evaluation instrument comprehensively covers all relevant aspects of the system's usability, reliability, performance, and accuracy.

Weighted Mean

The numerical ratings collected through the four-point Likert scale instrument are analyzed using the weighted mean formula to produce a single representative score for each evaluation item and for each functional module as a whole. The weighted mean is computed by multiplying the frequency of each rating value by its corresponding scale weight, summing the resulting products, and dividing the total by the number of respondents. This method ensures that the full distribution of responses across all four rating levels is accurately reflected in the final score, rather than relying on a simple average that may obscure variation in the respondent's opinion across different system features. The resulting weighted mean scores for each functional module are interpreted against the following scale: a score of 3.50 to 4.00 is interpreted as Strongly Agree or Outstanding; 2.50 to 3.49 as Agree or Satisfactory; 1.50 to 2.49 as Disagree or Needs Improvement; and 1.00 to 1.49 as Strongly Disagree or Unsatisfactory. These interpretation ranges provide a consistent and transparent basis for evaluating whether PrepAPig meets acceptable quality thresholds across all assessed dimensions. These dimensions include accuracy of growth trend calculations, response time for dashboard loading, reliability of offline synchronization, and user satisfaction with vaccination reminders. The scale also helps identify specific features that may require refinement before the system is considered fully operational for ANCY's Pig Farm.

Descriptive Analysis

In addition to the Likert scale and weighted mean computations, descriptive analysis is applied to summarize and interpret the overall evaluation results in a manner that is meaningful for the study's conclusions. Descriptive analysis involves the computation and reporting of frequency distributions, percentage scores, and overall mean values across all evaluation dimensions, providing a comprehensive picture of how the farm owner perceives PrepAPig's quality and the degree to which the system addresses the operational gaps identified in the current manual record-keeping process. For the AI-assisted features specifically, which include growth trend analysis and feed forecasting, descriptive analysis is also applied to compare the system's computed outputs against manually calculated values derived from the farm's historical records. This demonstrates that the system produces accurate, interpretable, and decision-relevant outputs. This dual application of descriptive analysis, encompassing both the evaluation instrument results and the system's analytical outputs, confirms that PrepAPig fulfills its intended function not only as a usable and reliable record-keeping platform but also as a genuinely data-driven decision support tool for the farm owner. Together, these three evaluation methods, namely the four-point Likert scale for standardized data collection, the weighted mean for quantitative scoring, and descriptive analysis for contextual interpretation, provide a rigorous and replicable framework for assessing whether PrepAPig fulfills its intended purpose as a functional, usable, AI-enabled, and offline-capable progressive web application for pig farm management at ANCY's Pig Farm.

3.9 Ethical Considerations

This section discusses the ethical responsibilities observed by the research team throughout the design, development, testing, and evaluation of PrepAPig. Given that the system collects farm records including pig batch information, growth weights, feed consumption logs, vaccination schedules, and capital expenses, and integrates artificial intelligence for growth trend analysis and feed forecasting, the researchers recognize that ethical conduct is not merely procedural but fundamental to the integrity and trustworthiness of the system and the study as a whole. The ethical considerations addressed in this section cover three primary areas: data privacy, client consent, and the responsible use of artificial intelligence.

Data Privacy

The study ensures strict adherence to data privacy in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). Only the minimum data required for PrepAPig's core functions, including pig growth monitoring, feed management, vaccination scheduling, and expense tracking, will be collected and utilized. All information will be processed responsibly, ensuring that no manipulation or unauthorized alteration of data will occur. In line with the Act, the researchers will safeguard all collected data and ensure that it will not be used beyond the scope of the study. All data transmissions will be encrypted, access to stored records will be restricted solely to the authenticated farm owner, and no farm records will be shared with external services except for non-personal notification payloads delivered through Firebase Cloud Messaging. This is to prevent any compromise to the confidentiality and operations of ANCY'S Pig Farm, while maintaining the integrity and security of all farm-related information.

Client Consent

Formal consent was obtained from the farm owner of ANCY's Pig Farm, who approved the conduct of this study through an official agreement. This approval confirms the owner's full understanding and authorization of the system's development and its intended application within the farm. The owner also expressed willingness to participate in the study, particularly in usability testing and system evaluation, to help ensure that PrepAPig aligns with actual farm needs. All other participants involved in testing are informed of the evaluation's nature beforehand, their participation is entirely voluntary, and they may withdraw at any time without consequence.

Responsible Use of Artificial Intelligence

The researchers will ensure that the use of artificial intelligence in PrepAPig is centered on ethical, transparent, and purpose-driven applications. AI will be primarily utilized for data analysis, allowing the system to process historical pig weight records, feed consumption logs, and vaccination schedules efficiently. Through this process, the system will generate growth trend charts and feed requirement forecasts, helping the owner identify patterns and trends relevant to pig development and farm productivity. The AI models are pre-defined and do not continuously retrain on live farm data, so all outputs remain clearly labeled as estimates based on past records. Furthermore, PrepAPig is designed strictly as a decision support tool, ensuring that final decisions on feeding, vaccination, and overall farm management will always be made by the farm owner.

In conclusion, this study adheres to ethical standards in developing PrepAPig by ensuring compliance with the Data Privacy Act of 2012, securing informed consent from

the farm owner of ANCY'S Pig Farm, and promoting the responsible use of AI for data analysis and decision support.

REFERENCES

- Aquilani, C., Confessore, A., Bozzi, R., Sirtori, F., & Pugliese, C. (2022). Review: Precision livestock farming technologies in pasture-based livestock systems. *Animal*, 16(1), 100429.
- Basumatary, B., & Agnihotri, N. (2022). Benefits and challenges of using NodeJS. *International Journal of Innovative Research in Computer Science & Technology*, 67-70.
- Berckmans, Daniel. "Why Precision Livestock Farming Can Generate a More Sustainable Livestock Sector." *and Welfare* (2021).
- Collins, L., & Smith, L. (2022). Review: Smart agri-systems for the pig industry. *animal*, 16, 100518. <https://doi.org/10.1016/j.animal.2022.100518>
- Damian Kostrzewa, & Marek Miłosz. (2024). Comparative analysis of the Express.js and ElysiaJS frameworks in the context of web application development. *Journal of Computer Sciences Institute*, 32, 246-250.
- Damian Kostrzewa, & Marek Miłosz. (2024). Comparative analysis of the Express.js and ElysiaJS frameworks in the context of web application development. *Journal of Computer Sciences Institute*, 32, 246-250.
- Distante, D., Albanello, C., Zaffar, H., Faralli, S., & Amalfitano, D. (2025). Artificial intelligence applied to precision livestock farming: A tertiary study. *Smart Agricultural Technology*, 11, 100889.
- Dwi Purnomo Putro, Adhika Pramita Widyassari, & Dea Salsabilla. (2023). Collaboration of progressive web app (PWA) and Firebase cloud messaging (FCM) for optimal

- performance mailing software. *INTERNATIONAL CONFERENCE ON DIGITAL ADVANCE TOURISM, MANAGEMENT AND TECHNOLOGY*, 1(2), 586-596.
- Funk, T., Rohrer, G., Brown-Brandl, T., & Keel, B. (2024). Online feeding behavior monitoring of individual group-housed grow-finish pigs using a low-frequency RFID electronic feeding system. *Translational Animal Science*, 8.
- Gómez, Y., Stygar, A. H., Boumans, I. J., Bokkers, E. A., Pedersen, L. J., Niemi, J. K., Pastell, M., Manteca, X., & Llonch, P. (2021). A systematic review on validated precision livestock farming technologies for pig production and its potential to assess animal welfare. *Frontiers in Veterinary Science*, 8.
- Hasanah, O. P. R. N. (2024). *Development of an independent library using HTML5 to improve students' digital literacy at vocational school*. Mount Hope Economic Global Journal (MEGA), 2(2).
- Jiddere, J. J., Anyachebelu, K. T., & Olalere, M. (2025). Integrating QR code authentication with Firebase cloud messaging: A performance-optimized architecture for enhanced mobile security and user experience. *Journal of Computer Allied Intelligence*, 3(5), 72-79.
- Koley, S., Singh, S., Ojha, B. K., Kurechiya, N., & Singh, A. (2022). Precision feeding in pig production: A review. *INDIAN JOURNAL OF ANIMAL HEALTH*.
- Komperla, V., Pratiba, D., Ghuli, P., & Pattar, R. (2021). React: A detailed survey. *Indonesian Journal of Electrical Engineering and Computer Science*, 26(3), 1710.

- Leshchuk, S. O., Ramskyi, Y. S., Kotyk, A. V., & Kutsiy, S. V. (2022). *Design a progressive web application to support student learning*. CEUR Workshop Proceedings.
- Marić, K., Gvozdanović, K., Djurkin Kušec, I., Kušec, G., & Margeta, V. (2025). Smart pig farms: Integration and application of digital technologies in pig production. *Agriculture*, 15(9), 937.
- Melak, A., Aseged, T., & Shitaw, T. (2024). The influence of artificial intelligence technology on the management of livestock farms. *International Journal of Distributed Sensor Networks*, 2024, 1-12.
- Mikita Piastou. (2023). Comprehensive performance and scalability assessment of front-end frameworks: React, angular, and Vue.js. *World Journal of Advanced Engineering Technology and Sciences*, 9(1), 366-376.
<https://doi.org/10.30574/wjaets.2023.9.2.0153>
- Miłosierny, B., & Dzieńkowski, M. (2021). The comparative analysis of web applications frameworks in the Node.js ecosystem. *Journal of Computer Sciences Institute*, 18, 42-48.
- Misiura, M. M., Filipe, J. A., & Kyriazakis, I. (2021). A novel estimation of unobserved pig growth traits for the purposes of precision feeding methods. *Frontiers in Veterinary Science*, 8.
- Neethirajan, S., & Kemp, B. (2021). Digital Phenotyping in livestock farming. *Animals*, 11(7), 2009. <https://doi.org/10.3390/ani11072009>

- Pham, L. M., Nguyen-Ba, H., Nguyen, H. S., & Le, H. (2021). Simulation of precision feeding systems for swine. *2021 13th International Conference on Knowledge and Systems Engineering (KSE)*, 1-6.
- Pires, J. G. (2023). Machine learning in medicine using JavaScript: Building web apps using TensorFlow.js for interpreting biomedical datasets.
- Raju Cherukuri, B. (2024). Progressive web apps (PWAs): Enhancing user experience through modern web development. *International Journal of Science and Research (IJSR)*, 13(10), 1550-1560. <https://doi.org/10.21275/ms241022095359>
- Salmi, H. A. (2023). Comparative CSS frameworks. *Multi-Knowledge Electronic Comprehensive Journal for Education and Science Publications(MECSJ)*, (66).
- Tangka, J. K., EVINA M. N., & Ngah, J. N. (2021). Development of a computerized automated system for feed, water and sanitation management in animal farms. *Current Journal of Applied Science and Technology*, 1-15.
- Van Klompenburg, T., & Kassahun, A. (2022). Data-driven decision making in pig farming: A review of the literature. *Livestock Science*, 261, 104961.,
- Wang, S., Jiang, H., Qiao, Y., Jiang, S., Lin, H., & Sun, Q. (2022). The research progress of vision-based artificial intelligence in smart pig farming. *Sensors*, 22(17), 6541. <https://doi.org/10.3390/s22176541>
- Zaveri, A. A., Jaafar, J., Yafi, E., & Sarama, S. (2024). Evolution of requirements engineering in Agile methodology – Literature review. *Journal of Engineering Technology and Applied Physics*, 6(2), 57-65. <https://doi.org/10.33093/jetap.2024.6.2.9>